Purely Analytic Supplement for
A Dirichlet Pólya Inequality on
Three-Dimensional Spherical Shells

16 July 2026

Abstract

This support volume contains the detailed hand proofs and exact-rational tables used to remove every executable predicate from the logical dependency chain of the spherical-shell theorem. Parts I–VI replace the 611-row exact ledger; Parts VII–IX prove the retained-remainder theorem for every frequency above the final staircase; and Part X assembles the final residual closure. The former aggregate-tail argument, programs, and interval certificates remain available only as optional independent regression checks and are not premises of this analytic proof.

Reading convention. The volume is modular so that each bottleneck lemma can be reviewed and hashed independently. Every included part is generated from its corresponding LaTeX source in the `manuscript/analytic` directory. The repeated title pages deliberately mark the trust boundary between exhibits.

Global census of the exact ledger

The canonical ledger has 553 core rows and 58 supplemental rows. The following inclusive ranges give a one-to-one allocation; the part numbers refer to this volume. The family codes denote, in order, rational-π checks (PI), thresholds (TH), strict-floor conventions (SF), zero specializations (Z), lower inventories (LI), checkpoint inventories (KI), cap payments (CP), localizations (LOC), the upper owner (U), optical screens (OPT), exact subtraction (SUB), and supplemental seam/payment rows (SUP).

canonical family	row range	unique owner
PI	PI1–PI4	II
TH	TH1–TH7, TH31–TH34 / TH8–TH30	I / II
SF	SF1–SF35	II
Z	Z1–Z48	I
LI	LI1–LI8, LI10–LI44 / LI9	I / II
KI	KI1–KI117	II
CP	CP1–CP43, CP56, CP60–CP77 / remaining rows	II / III
LOC	LOC1–LOC47	III
U	U1–U8	V
OPT	OPT1–OPT14 / OPT15–OPT50	V / II
SUB	SUB1–SUB68	VI
SUP	SUP1–SUP57 / SUP58	IV / I

Here "remaining rows" in CP means CP44–CP55, CP57–CP59, and CP78–CP98. Thus the six disjoint owner totals are

$$103 + 278 + 83 + 57 + 22 + 68 = 611.$$

In particular, the ten lower-d identities SUP36–SUP45 are assigned to Part IV although Part II also proves their general odd-sum identity; SUP58 is assigned to Part I; and OPT15–OPT50 are assigned to Part II although Part V reuses their consequences. These are proof redundancies, not row overlaps. The displayed ranges exhaust every canonical row and have empty pairwise intersections.

Contents

Hand-visible analytic derivations for 103 finite-ledger predicates

1 Scope and logical convention

This note exposes, in hand-checkable form, the predicates not assigned to the other five ledger packets. They are precisely

$$11 \text{ threshold predicates} + 48 \text{ zero predicates} + 43 \text{ lower-closure predicates} + 1 \text{ cap–74 predicate} = 103.$$

The omitted lower-closure predicate is the single six-cap tuple identity; it belongs to the cap-inventory packet. Every inequality below is proved from a displayed analytic inequality, integer arithmetic, rational cross multiplication, or positive-side squaring. An executable ledger may check the same arithmetic, but no executable output is used as a premise.

We use

$$\omega_0 = \frac{\sqrt{3}}{2\pi} - \frac{1}{6}, \quad C_0 = 1 + \frac{8\sqrt{2}}{15}, \quad \rho_* = \frac{\omega_0}{2C_0}, \quad \rho_0 = \frac{1}{\sqrt{337}}, \quad \sigma = \frac{3\omega_0}{4}, \quad \rho_c = \frac{1}{1+2\pi}, \quad d = \frac{\sqrt{337}}{2}.$$

The elementary enclosure

$$\frac{333}{106} < \pi < \frac{355}{113} < \frac{22}{7} \tag{1}$$

is analytic: the lower and middle bounds follow from alternating truncations of Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$, and the last one is immediate by cross multiplication. For reference, $355/113$ exceeds the upper Machin truncation by the positive rational numerator 45167474711 over denominator 189820334689453125. The classical integral

$$0 < \int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx = \frac{22}{7} - \pi$$

also gives the final upper bound directly.

2 The eleven threshold predicates

First we obtain a rational two-sided enclosure for ω_0. The exact integer residuals

$$3\,1101^2 113^2 - 607^2 355^2 = 1998482 > 0, \qquad 25\,333^2 - 243\,106^2 = 41877 > 0$$

and (1) imply

$$\sqrt{3} > \frac{607\,355}{1101\,113} > \frac{607\pi}{1101}, \qquad 9\sqrt{3} < 5\pi.$$

Consequently

$$\frac{40}{367} < \omega_0 < \frac{1}{9}. \tag{2}$$

Indeed, $607/2202 - 1/6 = 40/367$, while $5/18 - 1/6 = 1/9$.

Since $2\,32^2 - 45^2 = 23 > 0$, one has $\sqrt{2} > 45/32$, hence $C_0 > 7/4$ and $\rho_* < 2/63$. The following chain consists entirely of strict rational comparisons, with the indicated radical comparisons justified by squaring:

$$\rho_* < \frac{2}{63} < \frac{1}{\sqrt{337}} < \frac{1}{18} < \frac{3}{40} < \frac{3\omega_0}{4} < \frac{1}{12} < \frac{7}{51} < \frac{1}{1+2\pi} < \frac{1}{7} < \frac{39}{50} < \frac{7}{8}.$$

Here

$$4 \cdot 337 < 63^2, qquad 18^2 < 337, qquad 367 < 400, \qquad 51 < 84, qquad 367 < 420,$$

and $7/51 < \rho_c < 1/7$ follows respectively from $\pi < 22/7$ and $\pi > 3$. This proves the five ratio-order rows. It also proves

$$\omega_0 < \frac{1}{9} < \frac{7}{51} < \rho_c.$$

Moreover

$$\omega_0 d > \frac{20\sqrt{337}}{367} > 1, \qquad 400 \cdot 337 - 367^2 = 111 > 0.$$

The four terminal comparisons follow from the same bounds:

$$\sqrt{114} < 18 < \frac{2}{\omega_0}, \qquad \frac{2}{\omega_0} < \frac{367}{20},$$

$$\rho_c < \frac{1}{7} < \frac{60}{367} < \frac{3\omega_0}{2}, \qquad \omega_0\sqrt{114} > \frac{40\sqrt{114}}{367} > 1.$$

The last positive-side squaring has residual

$$1600 \cdot 114 - 367^2 = 47711 > 0.$$

Thus the one-to-one threshold registry is

ID	source row	proved predicate
T1–T5	1–5	$\rho_* < \rho_0 < \sigma < \rho_c < 39/50 < 7/8$, one adjacent pair per row
T6	6	$\omega_0 < \rho_c$
T7	7	$\omega_0 d > 1$
T8	31	$\sqrt{114} < 2/\omega_0$
T9	32	$2/\omega_0 < 367/20$
T10	33	$\rho_c < 3\omega_0/2$
T11	34	$\omega_0\sqrt{114} > 1$

3 The forty-eight zero specializations

3.1 A recurrence and elementary tangent bounds

Put

$$\mathrm{j}_\ell(x) = \sqrt{\frac{\pi}{2x}} J_{\ell+1/2}(x), \qquad P_\ell(x) = x^{\ell+1}\mathrm{j}_\ell(x).$$

Then

$$P_{\ell+1} = (2\ell+1)P_\ell - x^2 P_{\ell-1}, \qquad P_0 = \sin x, quad P_1 = \sin x - x\cos x. \tag{3}$$

Write $P_\ell = A_\ell(t)\sin x + xC_\ell(t)\cos x$, $t = x^2$. The recurrence gives

$$
\begin{aligned}
A_2 &= 3 - t, & C_2 &= -3,\\
A_3 &= 15 - 6t, & C_3 &= t - 15,\\
A_5 &= 945 - 420t + 15t^2, & C_5 &= -945 + 105t - t^2,\\
A_6 &= 10395 - 4725t + 210t^2 - t^3, & C_6 &= -10395 + 1260t - 21t^2,\\
A_7 &= 135135 - 62370t + 3150t^2 - 28t^3, & C_7 &= -135135 + 17325t - 378t^2 + t^3.
\end{aligned}
$$

For $0 < s < \pi/2$, differentiation and integration give

$$\tan s > s, qquad \tan s > s + \frac{s^3}{3}. \tag{4}$$

For $0 < s < 1$, $\sin s < s$ and $\cos s > 1 - s^2/2$ give

$$\tan s < \frac{s}{1 - s^2/2}. \tag{5}$$

3.2 Sixteen half-period locations

For each internal face, the first displayed margin below is $w-(h_L/2)(22/7)$; the second is $(h_R/2)(333/106)-w$. Positivity therefore proves both strict half-period location rows without a decimal approximation.

(ℓ, n)	w	(h_L, h_R)	left margin	right margin
$(1,2)$	$77/10$	$(4,5)$	$99/70$	$163/1060$
$(2,2)$	9	$(5,6)$	$8/7$	$45/106$
$(3,2)$	$103/10$	$(6,7)$	$61/70$	$737/1060$
$(5,2)$	$129/10$	$(8,9)$	$23/70$	$1311/1060$
$(1,3)$	$21/2$	$(6,7)$	$15/14$	$105/212$
$(2,3)$	$61/5$	$(7,8)$	$6/5$	$97/265$
$(6,1)$	10	$(6,7)$	$4/7$	$211/212$
$(7,1)$	$23/2$	$(7,8)$	$1/2$	$113/106$

3.3 Sixteen half-period signs

At an integral or half-integral multiple of π, exactly one of $\sin x, \cos x$ vanishes. Direct substitution in the displayed polynomials gives the following complete sign table.

(ℓ, n)	h_L	left sign	(ℓ, n)	h_R	right sign
$(1,2)$	4	$-$	$(1,2)$	5	$+$
$(2,2)$	5	$-$	$(2,2)$	6	$+$
$(3,2)$	6	$-$	$(3,2)$	7	$+$
$(5,2)$	8	$-$	$(5,2)$	9	$+$
$(1,3)$	6	$+$	$(1,3)$	7	$-$
$(2,3)$	7	$+$	$(2,3)$	8	$-$
$(6,1)$	6	$+$	$(6,1)$	7	$-$
$(7,1)$	7	$+$	$(7,1)$	8	$-$

Here are explicit polynomial witnesses for the only signs not immediate from $A_1 = 1, C_1 = -1, A_2 = 3 - t, C_2 = -3, A_3 = 15 - 6t, C_3 = t - 15$. From (1), at $x = 4\pi$ one has $144 < t < 159$, at $x = 9\pi/2$ one has $t > 182$, and at $x = 7\pi/2$ one has $110 < t < 121$. Monotonicity on these rational intervals and direct evaluation give

$$C_5(t) < C_5(144) = -6561, qquad A_5(t) > A_5(182) = 421365,$$
$$C_6(t) < C_6(81) = -46116, qquad A_6(t) > A_6(110) = 700645,$$
$$A_7(t) < A_7(110) = -5878565, qquad C_7(t) < C_7(144) = -2492559.$$

For example, $A_6' = -4725 + 420t - 3t^2$ is positive on $[110, 121]$, while A_7' is negative there; the other asserted monotonicities follow at once from their displayed derivatives. Combining these signs with the quadrant signs of sine and cosine proves all sixteen entries.

3.4 Eight rational-wall signs

Substitution in (3) reduces each rational-wall sign to one of the following exact tangent comparisons. The shifted angle is always in $(0, \pi/2)$, as already proved by the location table.

(ℓ, n)	wall	hand certificate	$\operatorname{sgn} \mathrm{j}_\ell(w)$
$(1,2)$	$77/10$	For $u = 5\pi/2 - w$, $\tan u > 3/20 > 10/77 = 1/w$; hence $\tan(w - 2\pi) = \cot u < w$.	$-$
$(2,2)$	9	For $u = 3\pi - w$, $\tan u > 21/50 > 9/26 = 3w/(w^2 - 3)$.	$-$
$(3,2)$	$103/10$	For $u = 7\pi/2 - w$, $\tan u > 69/100 > 621540/938227 = (6w^2 - 15)/(w(w^2 - 15))$.	$-$
$(5,2)$	$129/10$	For $y = w - 4\pi$, $0 < y < 17/50$, so $\tan y < 1700/4711 < 177800829/427700150 = -wC_5/A_5$.	$-$
$(1,3)$	$21/2$	For $u = 7\pi/2 - w$, $\tan u > 49/100 > 2/21 = 1/w$, hence $\tan(w - 3\pi) < w$.	$+$
$(2,3)$	$61/5$	For $u = 4\pi - w$, $\tan u > 9/25 > 915/3646 = 3w/(w^2 - 3)$.	$+$
$(6,1)$	10	For $y = w - 3\pi$, $4/7 < y < 29/50$, so $\tan y < 2900/4159 < 188790/127579 = -wC_6/A_6$.	$+$
$(7,1)$	$23/2$	For $u = 4\pi - w$, $\tan u > u + u^3/3 > 546377/375000 > 3153151489/2276518664 = wC_7/A_7$.	$+$

For complete cross-multiplication visibility, the four less immediate positive gaps in that table are

$$\frac{69}{100} - \frac{621540}{938227} = \frac{2583663}{93822700},$$
$$\frac{177800829}{427700150} - \frac{1700}{4711} = \frac{110529450419}{2014895406650},$$
$$\frac{188790}{127579} - \frac{2900}{4159} = \frac{415198510}{530601061},$$
$$\frac{546377}{375000} - \frac{3153151489}{2276518664} = \frac{3837851856583}{53355906187500}.$$

The four small gaps are respectively $31/1540, 24/325, 829/2100, 9939/91150$.

3.5 Six Lorch rows and two wall-order rows

For $\nu = \ell + 1/2$, Lorch's strict analytic first-zero estimate is

$$j_{\nu,1}^2 > \frac{24(\nu+1)^2}{1 - 2\nu + \sqrt{(2\nu+3)(2\nu+11)}} - 2(\nu^2 - 1) =: R_\ell.$$

After rationalization,

$$R_\ell = \frac{3(2\ell+3)\sqrt{(\ell+2)(\ell+6)} - 2\ell^2 + \ell + 6}{4}.$$

All sides being squared are positive, and the following integer residuals prove the three required strict specializations:

ℓ	wall c	positive squared residual proving $R_\ell > c^2$
8	71/6	$1026^2 \cdot 35 - 6067^2 = 35171$
9	64/5	$1575^2 \cdot 165 - 20059^2 = 6939644$
10	69/5	$3450^2 \cdot 3 - 5911^2 = 767579$

Since $\mathrm{j}_\ell(x) > 0$ for sufficiently small $x > 0$, and its first positive zero is strictly to the right of c, these same three rows imply $\mathrm{j}_\ell(c) > 0$. Thus the three executable sign rows are analytic corollaries, not independent numerical assumptions. Finally,

$$\frac{103}{10} - \frac{81}{8} = \frac{7}{40} > 0, \qquad \frac{61}{5} - \frac{21}{2} = \frac{17}{10} > 0.$$

The one-to-one zero registry is as follows. Number the eight internal faces in the order of the location table. For face r, IDs $Z_{5r-4}, \ldots, Z_{5r}$ are, in order: right of the left half-period, left of the right half-period, left half-period sign, right half-period sign, and rational-wall sign. This bijects IDs $Z1$–$Z40$ with the first forty source rows. The remaining bijection is

ID	source row	predicate
Z41	41	Lorch specialization $\ell = 8, c = 71/6$
Z42	42	positive sign at $\ell = 8, c = 71/6$
Z43	43	Lorch specialization $\ell = 9, c = 64/5$
Z44	44	positive sign at $\ell = 9, c = 64/5$
Z45	45	Lorch specialization $\ell = 10, c = 69/5$
Z46	46	positive sign at $\ell = 10, c = 69/5$
Z47	47	$103/10 > 81/8$
Z48	48	$61/5 > 21/2$

4 The forty-three lower-closure predicates

4.1 Eight activation and exclusion rows

The complete arithmetic is

$$\begin{aligned} (23/2)^2 - 114 &= 73/4 > 0, & 10^2 &= 100, \\ (129/10)^2 - 114 &= 5241/100 > 0, & 103/10 - 81/8 &= 7/40 > 0, \\ (81/8)^2 + 8 &= 7073/64, & (81/8)^2 + 18 - 114 &= 417/64 > 0, \\ (21/2)^2 + 4 - 114 &= 1/4 > 0, & 16\pi^2 &> 16 \cdot 3^2 > 114. \end{aligned}$$

These are exactly lower IDs L1–L8, corresponding to source rows 1–8.

4.2 Five Weyl payments

For $0 < \rho < \rho_c < 1/7$, (1) yields the uniform analytic lower bound

$$W(\rho, K) = \frac{2}{9\pi}(1 - \rho^3)K^3 > \frac{38}{539}K^3. \tag{6}$$

Using $d > 9$ and $\sqrt{7073}/8 > 21/2$, the five payments are the following exact positive reserves:

ID	source row	threshold/cap	reserve from (6)
L9	10	$K > d,\ 46$	$(38/539)9^3 - 46 = 2908/539$
L10	11	$K > 10,\ 59$	$(38/539)10^3 - 59 = 6199/539$
L11	12	$K > 81/8,\ 66$	$(38/539)(81/8)^3 - 66 = 990435/137984$
L12	13	$K > 21/2,\ 69$	$(38/539)(21/2)^3 - 69 = 555/44$
L13	14	$K > \sqrt{7073}/8,\ 78$	$(38/539)(21/2)^3 - 78 = 159/44$

4.3 The cap–395 cell: eighteen rows

Put $R = 367/20$, $z_\ell = \ell + 1/2$. The analytic turning estimate

$$G_K(z) < \frac{(K-z)^{3/2}}{3\sqrt{K}}$$

is increasing in K, so on $K < R$ a strict cap c_ℓ follows from

$$\frac{(R-z_\ell)^3}{9R} = \frac{(357-20\ell)^3}{1321200} < \left(c_\ell + \frac{3}{4}\right)^2. \tag{7}$$

The following table gives all fifteen caps and the exact positive numerator $82575(4c_\ell+3)^2 - (357-20\ell)^3$ obtained by clearing the positive denominators in (7).

ℓ	0	1	2	3	4	5	6	7
c_ℓ	6	5	5	4	4	3	3	3
margin	14697882	5409422	11827162	3611502	8555642	1604782	5267322	8361062

ℓ	8	9	10	11	12	13	14
c_ℓ	2	2	1	1	1	1	0
margin	2346202	4446342	176282	1474822	2444562	3133502	286642

Thus L15–L29 correspond bijectively to source rows 16–30, with L$(15+\ell)$ the ℓ-th table column. Since $G_R(z)$ is strictly decreasing in z, the $c_{14} = 0$ row removes every $\ell \geq 14$ tail; this is L30/source row 31.

The weighted cap itself, L14/source row 15, is visible from the contributions

$$6, 15, 25, 28, 36, 33, 39, 45, 34, 38, 21, 23, 25, 27, \qquad \sum_{\ell=0}^{13}(2\ell+1)c_\ell = 395.$$

Finally, on the exceptional cell $\rho K \geq 5/2$, with $\rho < 11/80$ and $1/\pi > 113/355$,

$$\begin{aligned} W(\rho, K) &> \frac{2}{9}\frac{113}{355}\left(\frac{5}{2}\right)^3\left[\left(\frac{80}{11}\right)^3 - 1\right] \\ &= \frac{160293325}{378004} = 395 + \frac{10981745}{378004} > 395. \end{aligned}$$

This is L31/source row 32.

4.4 Twelve floor and boundary-owner rows

The nine finite arithmetic rows are instances of one identity:

$$\frac{3}{2}(p+1) \leq 2p + \frac{1}{2} \quad (p \geq 2), qquad \left(2p + \frac{1}{2}\right) - \frac{3}{2}(p+1) = \frac{p-2}{2} \geq 0. \tag{8}$$

At $p = 2$, $(3/2)3 = 9/2$; this is L32/source row 33. For $p = 2, \ldots, 9$, formula (8) is respectively L33–L40/source rows 34–41.

The half-integer face is exact: $5/2 = 2 + 1/2$. Assigning it to the high side makes the interface remainder vanish and leaves the exceptional cell closed; this is L41/source row 42. At $K = 2/\omega_0$,

$$p = \lfloor \omega_0 K \rfloor = 2, \qquad \rho K \le \frac{2\rho_c}{\omega_0} < 3$$

by T10. This proves L42/source row 43. Moreover

$$\rho K - \frac{1}{2} < \frac{5}{2}, qquad m = \left\lfloor \rho K - \frac{1}{2} \right\rfloor \le 2 \le 2p - 1 = 3,$$

and the strict exceptional condition $K < 2/\omega_0$ fails at equality. Thus the equality face is aggregate-owned, proving L43/source row 44. More generally, if $K \ge 2/\omega_0$, then $p \ge 2$ and

$$\rho K < \frac{3}{2}(p+1) \le 2p + \frac{1}{2}, \qquad m \le 2p - 1,$$

so the finite instances conceal no unproved large-p case.

5 The supplemental cap–74 incompatibility

The propagated ball wall requires $z^2 > 2409/100$, while the product wall requires $z^2 < 70/3$. They are incompatible because

$$\frac{2409}{100} - \frac{70}{3} = \frac{7227 - 7000}{300} = \frac{227}{300} > 0.$$

This is supplemental ID S1 and is a hand proof, not merely a reported certificate value.

6 Census and source-visibility verdict

family	count	IDs	hand-visible status before this note
threshold remainder	11	T1–T11	printed, with a few implications compressed
zero specializations	48	Z1–Z48	tables printed; several recurrence steps compressed
lower remainder	43	L1–L43	payments printed; floor instances generated implicitly
supplemental cap–74	1	S1	printed exactly
total	103		

No predicate in this 103-row allocation is an unsupported executable-only mathematical premise. Before this note, the principal visibility gaps were the one-to-one expansion of the forty zero sign/location rows, the fact that the three Lorch-wall signs are logical corollaries of the strict first-zero bounds, and the expansion of the twelve floor/boundary rows. The displayed recurrences, rational margins, general floor identity, and exact ownership arguments close those presentation gaps.

Exact-ledger decomputerization:
Packets A and B

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Abstract

This note replaces two groups of executable exact-arithmetic checks in the three-dimensional spherical-shell Pólya proof. Packet A proves once and for all the strict-bracket convention, odd-multiplicity block sums, the optical product-deficit identity, and monotonicity of propagated zero and channel cuts. Packet B gives exact hand-checkable tables for the five moving $3z$-crossings and for the seven first-excluded checkpoint channels. It then proves, without a finite search, that the five printed checkpoint inventories are exhaustive. Every numerical comparison below is an equality of rational numbers, a cross multiplication, or a positive-side squaring. No program output, decimal approximation, or interval calculation is a premise.

Contents

1 Scope and notation

Throughout $0 < \rho < 1$. Put

$$z = z_\rho := \frac{\pi}{1-\rho}, \qquad k_m(\rho)^2 := z_\rho^2 + m(m+1),$$

and define the two ratio endpoints

$$\rho_0 := \frac{1}{\sqrt{337}}, \qquad \rho_c := \frac{1}{1+2\pi}.$$

The high-checkpoint argument uses the strict proxy

$$b_{\ell,n}(z)^2 := \max\{n^2z^2 + \ell(\ell+1),\, \beta_{\ell,n}^2\}. \tag{1}$$

A channel may be counted at K only when $b_{\ell,n}(z) < K$. The constants $\beta_{\ell,n}$ are strict lower bounds for the corresponding ball zeros. The zero bounds used in the checkpoint table are displayed explicitly in Section 5.

The purpose of this note is deliberately limited. It replaces the general exact-arithmetic and checkpoint-ordering rows of the old ledger; it does not replace the zero-sign proofs, high-event payment registry, seam arithmetic, optical endpoint screens, or formal owner subtraction.

2 A rational enclosure for π

Lemma 2.1 (Hand enclosure for π)**.** *One has*

$$\frac{333}{106} < \pi < \frac{22}{7}, \qquad \frac{1}{\pi} < \frac{106}{333}. \tag{2}$$

Proof. For $0 < x \le 1$, the alternating series for $\arctan x$ gives a strict lower bound after a negative term and a strict upper bound after a positive term. Hence

$$\arctan\frac{1}{5} > \frac{1}{5} - \frac{1}{3\cdot 5^3} + \frac{1}{5\cdot 5^5} - \frac{1}{7\cdot 5^7} = \frac{323852}{1640625}, \qquad \arctan\frac{1}{239} < \frac{1}{239}.$$

Machin's identity therefore yields

$$\pi = 16\arctan\frac{1}{5} - 4\arctan\frac{1}{239} > \frac{1231847548}{392109375},$$

and direct subtraction gives the positive margin

$$\frac{1231847548}{392109375} - \frac{333}{106} = \frac{3418213}{41563593750} > 0.$$

For the upper bound, polynomial division and elementary integration give

$$0 < \int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx = \frac{22}{7} - \pi.$$

Finally, reciprocation of the positive lower bound for π gives the last inequality. Notice also that $22/7 - 333/106 = 1/742 > 0$, so the enclosure is nonempty. □

3 Packet A: universal exact-arithmetic lemmas

3.1 Strict floors and equality owners

Definition 3.1. For $x \in \mathbb{R}$, let

$$[x]_< := \#\{q \in \mathbb{Z}_{\ge 1} : q < x\}.$$

Thus the bracket counts positive integers under a *strict* cutoff.

Lemma 3.2 (Strict-bracket identity)**.** *For every real x,*

$$[x]_< = \max\{0, \lceil x \rceil - 1\}. \tag{3}$$

In particular, for every integer $m \geq 1$,

$$[m]_< = m - 1, \qquad [m + \tfrac{1}{2}]_< = m.$$

Consequently an integer equality wall is owned by the excluded side.

Proof. If $x \leq 1$, both sides of (3) vanish. If $x > 1$, the positive integers below x are exactly $1, \ldots, \lceil x \rceil - 1$. This also proves the two specializations. $\square$

The same lemma handles the two equality conventions used later. If $A + 1/4 = n \in \mathbb{Z}_{\geq 1}$, then $[A + 1/4]_< = n - 1$. If $a = m\pi$ in the optical radial decomposition, the strict radial layers are represented by $(N, t) = (m - 1, 1)$, not by $(m, 0)$.

3.2 Odd multiplicities and block caps

Lemma 3.3 (Odd-sum and block-cap identity)**.** *For every integer $h \geq 0$,*

$$\sum_{\ell=0}^{h} (2\ell + 1) = (h + 1)^2, \qquad \sum_{\ell=0}^{h-1} (2\ell + 1) = h^2. \tag{4}$$

More generally, if radial layer n contains a complete angular block $0 \leq \ell \leq h_n$, with $h_n = -1$ denoting an empty block, then its total multiplicity is

$$\sum_n (h_n + 1)^2. \tag{5}$$

If $x^2 = h(h + 1)$ is a strict product equality wall, the equality channel $\ell = h$ is absent and the contribution is h^2.

Proof. The first identity follows either by induction or by $(\ell + 1)^2 - \ell^2 = 2\ell + 1$. Summing over independent radial layers gives (5). At $x^2 = h(h + 1)$, $\sqrt{x^2 + 1/4} - 1/2 = h$; Lemma 3.2 excludes $\ell = h$, leaving $0 \leq \ell < h$ and hence h^2. $\square$

For completeness, the cap arithmetic collapsed by this lemma is printed in compact form below. A dash means that another analytic argument has declared the cell impossible; the lemma asserts only the arithmetic of the non-dash entries.

$k_8 : H \backslash h$	$\varnothing$	0	1	2	3
7	64	65	68	73	80
6	49	50	53	–	–
5	36	37	40	45	52
4	25	26	29	34	41

$k_9 : H \backslash h$	$\varnothing$	0	1	2	3	4
8	81	–	–	–	–	–
7	64	65	68	73	–	–
6	49	50	53	58	65	74
5	36	37	40	45	52	61
4	25	26	29	34	41	50

Indeed every entry is $(H+1)^2$ plus 0 or $(h+1)^2$. At k_7, the first blocks $H = 4, 5, 6$ give $25, 36, 49$, and second channels $0, 1$ contribute $1 + 3 = 4$. The remaining printed caps decompose as

$$\begin{aligned} k_{10}: \quad & 10^2; \quad 9^2,\ 9^2+4^2; \\ & 8^2,\ 8^2+4^2,\ 8^2+5^2,\ 8^2+6^2; \\ & 7^2+5^2,\ 7^2+4^2+2^2,\ 7^2+5^2+2^2, 7^2+6^2,\ 7^2+6^2+2^2 \\ & = 100; \quad 81, 97; \quad 64, 80, 89, 100; \quad 74, 69, 78, 85, 89, \end{aligned}$$

$$\begin{aligned} k_{11}: \quad & 11^2; \quad 10^2,\ 10^2+5^2; \quad 9^2+5^2,\ 9^2+5^2+2^2; \quad 8^2+5^2+2^2 \\ & = 121; \quad 100, 125; \quad 106, 110; \quad 93. \end{aligned}$$

The conservative cap-74 row is the same identity $7^2 + 5^2 = 74$.

The six lower inventories similarly give, in their printed order,

$$(6^2+3^2,\ 6^2+3^2+1,\ 7^2+3^2+1, 7^2+4^2+1,\ 7^2+4^2+1+3,\ 7^2+5^2+1+3) = (45, 46, 59, 66, 69, 78).$$

Finally, the ten lower-d caps are the successive odd sums

channels	cap	channels	cap
1	1	$1+3$	4
$1+3+5$	9	$1+3+5+1$	10
$1+3+5+7$	16	$1+3+5+7+1$	17
$1+3+5+7+9+1$	26	$1+3+5+7+9+1+3$	29
$1+3+5+7+9+11+1+3$	40	$1+3+5+7+9+11+1+3+5$	45.

3.3 The symbolic optical product deficit

Proposition 3.4 (Product-deficit identity and uniform reserve)**.** *Let $N \geq 1$ be an integer, $0 < t \leq 1$, and $x = N + t$. Then*

$$\frac{2}{3}x^3 - Nx^2 + \sum_{n=1}^{N} n^2 = \frac{N^2}{2} + N\left(t^2 + \frac{1}{6}\right) + \frac{2t^3}{3}. \tag{6}$$

For $C_D = 1382/3125$, the right-hand side is strictly larger than $C_D(N+t)^2$. More precisely, its difference from $C_D(N+t)^2$ is at least

$$\frac{1822532}{91552734375} > 0. \tag{7}$$

Proof. Insert $\sum_{n=1}^{N} n^2 = N(N+1)(2N+1)/6$ and expand; this proves (6) for all N, t, rather than at sampled values. Put $C = C_D$, $A = 1/2 - C = 361/6250$, and let $G_N(t)$ denote the difference under consideration. Then

$$G_N(t) = AN^2 + N\left(t^2 - 2Ct + \frac{1}{6}\right) + \frac{2t^3}{3} - Ct^2.$$

Consequently

$$G_{N+1}(t) - G_N(t) = A(2N+1) + t^2 - 2Ct + \frac{1}{6}.$$

Since $0 < C < 1$, the last quadratic has minimum $1/6 - C^2$ on $(0, 1]$. For $N \geq 1$,

$$G_{N+1}(t) - G_N(t) \geq 3A + \frac{1}{6} - C^2 = \frac{4229603}{29296875} > 0.$$

It remains to minimize G_1. Direct differentiation gives

$$G_1'(t) = 2(t - C)(t + 1),$$

so its unique minimum on $0 < t \leq 1$ is at $t = C$. Exact substitution gives

$$G_1(C) = \frac{1822532}{91552734375} > 0,$$

which proves both assertions. $\square$

3.4 Monotonic propagated cuts

Lemma 3.5 (Zero-cut/channel-cut monotonicity)**.** *Fix $n \geq 2$, a checkpoint m, and $M = m(m+1)$. Let $\mathcal{A}_n$ be a finite collection of anchor pairs $(j, q_{j,n}^2)$, where $j_{j+1/2,n}^2 > q_{j,n}^2$ at each anchor. For every ℓ above at least one anchor, set*

$$\mathfrak{B}_n(\ell) := \max_{(j,q_{j,n}^2)\in\mathcal{A}_n,\ j\leq\ell} \big(q_{j,n}^2 + \ell(\ell+1) - j(j+1)\big), \tag{8}$$

$$\mathfrak{Z}_{m,n}(\ell) := \mathfrak{B}_n(\ell) - M, \tag{9}$$

$$\mathfrak{C}_{m,n}(\ell) := \frac{M - \ell(\ell+1)}{n^2 - 1}. \tag{10}$$

Then

$$\mathfrak{Z}_{m,n}(\ell+1) \geq \mathfrak{Z}_{m,n}(\ell) + 2(\ell+1) > \mathfrak{Z}_{m,n}(\ell), \tag{11}$$

$$\mathfrak{C}_{m,n}(\ell+1) = \mathfrak{C}_{m,n}(\ell) - \frac{2(\ell+1)}{n^2-1} < \mathfrak{C}_{m,n}(\ell). \tag{12}$$

In particular, if $\mathfrak{Z}_{m,n}(\ell_0) \geq \mathfrak{C}_{m,n}(\ell_0)$, then no channel (ℓ, n) with $\ell \geq \ell_0$ can satisfy the strict proxy (1) at $K = k_m$.

Proof. Every old candidate in (8) gains exactly $(\ell+1)(\ell+2) - \ell(\ell+1) = 2(\ell+1)$ when ℓ is replaced by $\ell + 1$; a new anchor can only increase the maximum. This proves (11). Formula (12) is direct.

If a channel were counted at k_m, its product part would give

$$(n^2 - 1)z^2 < M - \ell(\ell+1), \quad \text{hence} \quad z^2 < \mathfrak{C}_{m,n}(\ell),$$

whereas its propagated ball-zero part would give

$$\mathfrak{B}_n(\ell) < z^2 + M, \quad \text{hence} \quad z^2 > \mathfrak{Z}_{m,n}(\ell).$$

These strict inequalities are incompatible when $\mathfrak{Z} \geq \mathfrak{C}$. Equations (11) and (12) propagate the incompatibility to every larger angular index. $\square$

4 Packet B: moving-$3z$ ordering

We first record rational radical enclosures. All six follow by positive squaring; the exact square margins are

lower comparison	margin	upper comparison	margin
$(183/10)^2 < 337$	$211/100$	$337 < (92/5)^2$	$39/25$
$(1051/100)^2 < 7073/64$	$2221/40000$	$7073/64 < (1052/100)^2$	$6191/40000$
$(1067/100)^2 < 114$	$1511/10000$	$114 < (107/10)^2$	$49/100.$

Thus

$$\frac{183}{10} < \sqrt{337} < \frac{92}{5}, \quad \frac{1051}{100} < \frac{\sqrt{7073}}{8} < \frac{1052}{100}, \quad \frac{1067}{100} < \sqrt{114} < \frac{107}{10}. \tag{13}$$

Lemma 4.1 (Endpoint and fixed-wall order)**.** *With* $d = \sqrt{337}/2$*,*

$$d < 3z_{\rho_0} < 10 < \frac{81}{8} < \frac{21}{2} < \frac{\sqrt{7073}}{8} < \sqrt{114} < 3z_{\rho_c}. \tag{14}$$

Moreover $(81/8)^2 + 8 = 7073/64$.

Proof. The nontrivial endpoint estimates have the following rational margins. From (2) and (13),

$$3z_{\rho_0} > 3\pi > \frac{999}{106}, \qquad d < \frac{46}{5}, \qquad \frac{999}{106} - \frac{46}{5} = \frac{119}{530} > 0.$$

Also $\rho_0 < 10/183$, whence

$$3z_{\rho_0} < \frac{66}{7}\frac{183}{173} = \frac{12078}{1211}, \qquad 10 - \frac{12078}{1211} = \frac{32}{1211} > 0.$$

At the other endpoint, $z_{\rho_c} = \pi + 1/2$; therefore

$$3z_{\rho_c} > \frac{579}{53}, \qquad \sqrt{114} < \frac{107}{10}, \qquad \frac{579}{53} - \frac{107}{10} = \frac{119}{530} > 0.$$

The rational middle comparisons are immediate. For the two radical steps one may square positive quantities:

$$\frac{7073}{64} - \left(\frac{21}{2}\right)^2 = \frac{17}{64} > 0, \qquad 114 - \frac{7073}{64} = \frac{223}{64} > 0.$$

Finally, $(81/8)^2 + 8 = (6561 + 512)/64 = 7073/64$. □

For a positive wall w, the unique solution of $3z_\rho = w$ is

$$r_w := 1 - \frac{3\pi}{w}.$$

The following table encloses each crossing between explicit rational numbers. The bounds use only Lemma 2.1 and (13); for instance r_w increases with w and decreases with π.

wall w	exact rational enclosure for w	exact enclosure for r_w
10	10	$\frac{2}{35} < r_{10} < \frac{61}{1060}$
81/8	81/8	$\frac{13}{189} < r_{81/8} < \frac{11}{159}$
21/2	21/2	$\frac{5}{49} < r_{21/2} < \frac{38}{371}$
$\sqrt{7073}/8$	$1051/100 < w < 1052/100$	$\frac{757}{7357} < r_w < \frac{2903}{27878}$
$\sqrt{114}$	$1067/100 < w < 107/10$	$\frac{79}{679} < r_w < \frac{676}{5671}$

The four gaps between consecutive rational enclosures are, in order,

$$\frac{2251}{200340}, \qquad \frac{256}{7791}, \qquad \frac{183}{389921}, \qquad \frac{231225}{18929162}, \tag{15}$$

and are all positive. For example, the first is $13/189 - 61/1060$, and similarly for the other three. The endpoint margins are also explicit:

$$\rho_0 < \frac{10}{183} < \frac{2}{35}, \quad \frac{2}{35} - \frac{10}{183} = \frac{16}{6405} > 0,$$

and, since $\rho_c > 7/51$,

$$\frac{676}{5671} < \frac{7}{51} < \rho_c, \quad \frac{7}{51} - \frac{676}{5671} = \frac{5221}{289221} > 0.$$

We have therefore proved the complete strict order

$$\rho_0 < r_{10} < r_{81/8} < r_{21/2} < r_{\sqrt{7073}/8} < r_{\sqrt{114}} < \rho_c. \tag{16}$$

This single table replaces the ten crossing-location predicates and the four crossing-order predicates of the executable registry.

5 Packet B: checkpoint boundary table and exhaustiveness

The ball-zero input consists of the following strict rational lower bounds:

$$\begin{array}{c|cccc} n=2,\ \ell & 1 & 2 & 3 & 5 \\ \hline q^2_{\ell,2} & (77/10)^2 & 9^2 & (103/10)^2 & (129/10)^2 \end{array} \qquad \begin{array}{c|cc} n=3,\ \ell & 1 & 2 \\ \hline q^2_{\ell,3} & (21/2)^2 & (61/5)^2. \end{array} \tag{17}$$

Angular propagation gives the lower bound (8) at all larger angular indices. The seven first-excluded channels have the exact cuts shown below. Here $M = m(m+1)$, $B = \mathfrak{B}_n(\ell_0)$, $Z = B - M$, $C = (M - \ell_0(\ell_0+1))/(n^2-1)$, and the last column is the decisive incompatibility margin $Z - C$.

m	n	first excluded ℓ_0	B	Z	C	$Z-C$
7	2	2	81	25	50/3	25/3
8	2	4	11409/100	4209/100	52/3	7427/300
9	2	5	16641/100	7641/100	20	5641/100
10	2	6	17841/100	6841/100	68/3	13723/300
11	2	5	16641/100	3441/100	34	41/100
10	3	2	3721/25	971/25	13	646/25
11	3	2	3721/25	421/25	63/4	109/100

Every table entry is reconstructible from (17). The only propagated base values not already anchors are

$$\frac{11409}{100} = \left(\frac{103}{10}\right)^2 + (4\cdot 5 - 3\cdot 4), \qquad \frac{17841}{100} = \left(\frac{129}{10}\right)^2 + (6\cdot 7 - 5\cdot 6).$$

All seven margins are positive; the smallest is $41/100$, so none of these exclusions relies on an equality convention. By Lemma 3.5, each row excludes not merely the displayed channel but its entire angular tail.

It remains to rule out the radial tails. On the high strip $\rho_c \le \rho < 1$,

$$z \ge z_{\rho_c} = \pi + \frac{1}{2} > \frac{333}{106} + \frac{1}{2} = \frac{193}{53}, \qquad z^2 > \frac{37249}{2809}.$$

The exact margins needed below are

$$\frac{37249}{2809} - \frac{90}{8} = \frac{22591}{11236} > 0, \qquad \frac{37249}{2809} - \frac{132}{15} = \frac{62649}{14045} > 0. \tag{18}$$

At $m \le 9$, an $n = 3$ channel would require $8z^2 < M - \ell(\ell+1) \le 90$, contradicting the first inequality. At $m \le 11$, every $n \ge 4$ channel would require $15z^2 \le (n^2-1)z^2 < M - \ell(\ell+1) \le 132$, contradicting the second inequality. These are the two base radial exclusions.

Theorem 5.1 (Exhaustive checkpoint inventories)**.** *For $\rho_c \le \rho < 1$, the channels that can satisfy the strict proxy* (1) *at the five checkpoints are contained in the following lists:*

checkpoint	*possible modes*
k_7	*first* 0:6, *second* 0:1, *no* $n \ge 3$
k_8	*first* 0:7, *second* 0:3, *no* $n \ge 3$
k_9	*first* 0:8, *second* 0:4, *no* $n \ge 3$
k_{10}	*first* 0:9, *second* 0:5, *third* 0:1, *no* $n \ge 4$
k_{11}	*first* 0:10, *second* 0:4, *third* 0:1, *no* $n \ge 4$.

Here n-th means radial index n, and a:b denotes the angular indices $a, a+1, \dots, b$. The word "possible" asserts an exhaustive upper inventory, not that every listed channel is present for every ratio.

Proof. For $n = 1$, the product part of the strict proxy at k_m is

$$z^2 + \ell(\ell+1) < z^2 + m(m+1).$$

Thus $\ell < m$. The equality channel $\ell = m$ and all larger indices are excluded, with the equality ownership fixed by Lemma 3.2. This gives precisely first 0:$m-1$.

For $n = 2$, apply the first five rows of the checkpoint table and then Lemma 3.5. They give first-excluded angular indices $2, 4, 5, 6, 5$ at $m = 7, 8, 9, 10, 11$, respectively. Hence the second blocks end at $1, 3, 4, 5, 4$.

For $n = 3$, the first inequality in (18) removes all channels through k_9. At k_{10} and k_{11}, the last two rows of the checkpoint table and cut monotonicity remove every $\ell \ge 2$, leaving only $\ell = 0, 1$. Finally, the second inequality in (18) removes all $n \ge 4$ through k_{11}. The five lists follow, without any bounded angular search. ☐

6 Executable-row coverage contract

The former executable baseline contained 611 rows and had canonical digest

`afd7e054f52aa9a3992fc8ef16d3e7551586c76bfb589053714f5f55a36792f1`.

The digest is included only to identify the audited baseline. It is not a premise of any proof above. The exhibits above replace disjoint row groups as follows:

analytic exhibit	rows	exact ledger subfamilies
Lemma 2.1	4	one enclosure-nonemptiness row, two rational π-bound rows, and one reciprocal row
Lemma 3.2	24	sixteen integer/half-integer strict brackets, seven phase-proxy integer walls, and one optical radial equality owner
Lemma 3.3	97	twelve strict angular equality blocks, twelve optical angular blocks, 62 high-checkpoint cap identities, one six-cap lower tuple, and ten lower-d cap identities
Proposition 3.4	23	five general product-deficit predicates and eighteen sampled expansions
Lemma 3.5, the checkpoint boundary table, and Theorem 5.1	117	76 repeated cut-monotonicity rows, seven first-excluded comparisons, two radial exclusions, and 32 metadata/tail instances
Lemma 4.1 and the moving-crossing table	23	five fixed-wall orders, one propagation identity, three endpoint comparisons, ten crossing locations, and four crossing-order rows
total	**288**	

The following family table gives the same accounting against the executable entry points and makes the partial-family boundaries explicit. It maps the exact rows replaced by this note. "Collapsed" means that multiple sampled or bookkeeping rows are consequences of one displayed identity; it does not mean that an unprinted program enumeration is being used.

executable family	family rows	replaced here	analytic replacement
`verify_pi`	4	4	Lemma 2.1
`verify_thresholds`	34	23	five fixed-wall orders, one propagation identity, three endpoint comparisons, and fourteen moving-crossing predicates
`verify_strict_face_conventions`	35	35	Lemmas 3.2 and 3.3
`verify_lower_inventories_and_payments`	44	1	single six-cap tuple bookkeeping row
`verify_checkpoint_inventories`	117	117	boundary table, radial margins, and Theorem 5.1
`verify_cap_tables_and_payments`	98	62	all block-cap arithmetic: 61 cap-table rows plus the cap-74 block identity
`verify_optical_constants`	50	36	twelve angular blocks, one radial equality owner, five general product-deficit facts, and eighteen sampled expansions
supplemental k_6/lower-d ledger	26	10	ten lower-d block-cap identities
total		**288**	

For the checkpoint family, the 117 rows decompose exactly as follows: 32 finite metadata/tail instances, seven first-excluded zero/channel comparisons, two base radial exclusions, and 76 repeated cut-monotonicity instances. Theorem 5.1 replaces all four groups. For the optical family, Proposition 3.4 replaces the 18 sampled (N, t) expansions by a symbolic identity valid for every integer $N \geq 1$ and every $0 < t \leq 1$.

The remaining 323 rows are outside Packets A and B. In particular, this note makes no claim to replace the high-event cap/payment implications, the Bessel sign registry, seam payments, optical screen constants, or the exact owner subtraction. Those require the other analytic replacement packets. This separation prevents a universal arithmetic lemma from being mistaken for proof of a geometric event assignment.

Connected High-Event Registry

Analytic ledger replacement, Packet C

Abstract

This note replaces the exact-executable arithmetic used in the high spherical-shell staircase by displayed rational witnesses. It contains the complete seventeen-face localization registry, the three algebraic splits, the three dashed (impossible) cap cells, the five global checkpoint payments, the seven conditional payments, and the corrected cap–74 path. All comparisons are strict rational comparisons after the two elementary bounds $333/106 < \pi < 22/7$. In particular, no decimal approximation, interval package, or external ledger is a logical input to this packet.

1 Definitions and strict proxy convention

For $0 < \rho < 1$, put

$$z = z_\rho := \frac{\pi}{1-\rho}, \qquad x = x_\rho := z_\rho^2, \qquad k_m(\rho) := \sqrt{x_\rho + m(m+1)}. \tag{1}$$

The three-dimensional Weyl term is

$$\mathcal{W}(\rho, K) := \frac{2}{9\pi}(1-\rho^3)K^3. \tag{2}$$

At $K = k_m(\rho)$, the product part of the strict channel proxy for the mode (ℓ, n), $n \geq 2$, is

$$n^2 x + \ell(\ell+1) < x + m(m+1).$$

It is therefore governed by

$$\Delta_{m;n,\ell}(\rho) := m(m+1) - \ell(\ell+1) - (n^2-1)x_\rho. \tag{3}$$

The channel is product-possible only when $\Delta_{m;n,\ell} > 0$. Equality is excluded because the spectral proxy is strict. Since

$$x'_\rho = \frac{2\pi^2}{(1-\rho)^3} > 0, \qquad \Delta'_{m;n,\ell} = -(n^2-1)x'_\rho < 0, \tag{4}$$

a positive witness at r controls the side $\rho \leq r$, while a negative witness controls the side $\rho \geq r$.

We write H for the largest first-mode angular index retained in a cap cell and h for the largest second-mode angular index. A complete block through H has multiplicity $(H+1)^2$, because $\sum_{\ell=0}^{H}(2\ell+1) = (H+1)^2$.

2 The only transcendental enclosure

Lemma 1 (Rational enclosure of π)**.** *One has*

$$\pi_- := \frac{333}{106} < \pi < \frac{22}{7} =: \pi_+. \tag{5}$$

Proof. For $0 < u < 1$, the alternating expansion of $\arctan u$ lies between each pair of consecutive partial sums. The tangent addition formula gives

$$\tan\left(4\arctan\frac{1}{5} - \arctan\frac{1}{239}\right) = 1;$$

indeed, $\tan(4\arctan(1/5)) = 120/119$, so both the numerator and denominator in the last tangent quotient are $28561/(119 \cdot 239)$. The angle lies in $(0, \pi/2)$, and hence Machin's identity follows:

$$\pi = 16\arctan\frac{1}{5} - 4\arctan\frac{1}{239}.$$

Using an even partial sum for the positive term and an upper partial sum for the subtracted term gives

$$\begin{aligned} \pi &> 16\left(\frac{1}{5} - \frac{1}{3\cdot 5^3} + \frac{1}{5\cdot 5^5} - \frac{1}{7\cdot 5^7}\right) - \frac{4}{239}, \\ [\text{right side}] - \frac{333}{106} &= \frac{3418213}{41563593750} > 0. \end{aligned}$$

Likewise,

$$\begin{aligned} \pi &< 16\left(\frac{1}{5} - \frac{1}{3\cdot 5^3} + \frac{1}{5\cdot 5^5}\right) - 4\left(\frac{1}{239} - \frac{1}{3\cdot 239^3}\right), \\ \frac{22}{7} - [\text{right side}] &= \frac{1845738322}{1493178640625} > 0. \end{aligned}$$

This proves (5). $\square$

3 The seventeen rational localization faces

Let

$$\rho_c := \frac{1}{1+2\pi}.$$

Because $\pi > 3$, $\rho_c < 1/7$. The seventeen registry ratios are strictly ordered as follows:

$$\begin{aligned} &\frac{4}{25} < \frac{1}{5} < \frac{1}{4} < \frac{3}{10} < \frac{1}{3} < \frac{7}{20} < \frac{3}{8} < \frac{2}{5} < \frac{5}{12} < \frac{3}{7} < \frac{1}{2} \\ &\qquad < \frac{8}{15} < \frac{107}{200} < \frac{3}{5} < \frac{16}{25} < \frac{13}{20} < \frac{2}{3}. \end{aligned} \tag{6}$$

For a completely explicit distinctness check, the sixteen consecutive gaps in (6) are

$$\frac{1}{25}, \frac{1}{20}, \frac{1}{20}, \frac{1}{30}, \frac{1}{60}, \frac{1}{40}, \frac{1}{40}, \frac{1}{60}, \frac{1}{84}, \frac{1}{14}, \frac{1}{30}, \frac{1}{600}, \frac{13}{200}, \frac{1}{25}, \frac{1}{100}, \frac{1}{60}. \tag{7}$$

Moreover

$$\frac{4}{25} - \frac{1}{7} = \frac{3}{175} > 0, \qquad \frac{7}{8} - \frac{2}{3} = \frac{5}{24} > 0.$$

Thus all seventeen ratios lie in the high strip $\rho_c < \rho < 7/8$.

For a positive row below, we use

$$\Delta_{m;n,\ell}(r) > m(m+1) - \ell(\ell+1) - (n^2-1)\left(\frac{\pi_+}{1-r}\right)^2.$$

For a negative row, we use

$$-\Delta_{m;n,\ell}(r) > (n^2-1)\left(\frac{\pi_-}{1-r}\right)^2 - m(m+1) + \ell(\ell+1).$$

The resulting exact witnesses are the following. "Retain" means only that the product wall does not permit deletion of the indicated block; the independent Bessel wall may still remove it.

no.	r	$(m;n,\ell)$	exact sign witness	one-sided cap/event consequence
1	1/5	$(11;3,1)$	$\Delta > 320/49$	retain the third block 0:1 for $\rho \le 1/5$
2	3/10	$(8;2,3)$	$-\Delta > 58215/137641$	at k_8, $h \le 2$ for $\rho \ge 3/10$
3	1/3	$(8;2,2)$	$-\Delta > 27699/44944$	at k_8, $h \le 1$ for $\rho \ge 1/3$
4	3/8	$(9;2,3)$	$\Delta > 2622/1225$	retain the $h=3$ column below this face
5	2/5	$(9;2,2)$	$\Delta > 248/147$	retain the $h=2$ column below this face
6	5/12	$(9;2,1)$	$\Delta > 2200/2401$	retain the $h=1$ column below this face
7	3/7	$(9;2,0)$	$-\Delta >$ 120843/179776	no second mode at k_9 on the upper side
8	1/2	$(10;2,0)$	$-\Delta > 23677/2809$	no second mode at k_{10} on the upper side
9	107/200	$(11;2,0)$	$-\Delta >$ 13302732/2699449	no second mode at k_{11} on the upper side
10	8/15	$(11;2,0)$	$-\Delta >$ 2175627/550564	the clean stated cutoff: a second mode forces $\rho < 8/15$
11	3/5	$(8;2,0)$	$-\Delta >$ 5080707/44944	select the no-second k_8 cap/payment side
12	16/25	$(9;2,0)$	$-\Delta >$ 1555635/11236	select the no-second k_9 cap/payment side
13	13/20	$(10;2,0)$	$-\Delta >$ 18126190/137641	select the no-second k_{10} cap/payment side
14	2/3	$(11;2,0)$	$-\Delta >$ 1510851/11236	select the no-second k_{11} cap/payment side
15	4/25	$(10;3,0)$	$-\Delta >$ 260740/137641	no third mode at k_{10} on the upper side
16	1/4	$(11;3,0)$	$-\Delta > 23484/2809$	a third mode at k_{11} forces $\rho < 1/4$
17	7/20	$(9;2,4)$	$-\Delta > 36230/474721$	the $H=6, h=4$ cap–74 cell can occur only below this face

Every entry is a displayed positive rational number. Together with the monotonicity (4), the table proves the asserted side on an interval, not merely at a sampled point.

4 Three algebraic splits and the five k_{11} implications

At $\rho = \rho_c$, $x = (\pi + 1/2)^2$, while at $\rho = 7/8$, $x = 64\pi^2$. Lemma 1 gives the exact chain

$$(\pi + \tfrac{1}{2})^2 < \left(\frac{51}{14}\right)^2 = \frac{2601}{196} < 16 < \frac{68}{3} < 34 < 64 \cdot 3^2 < 64\pi^2. \tag{8}$$

Consequently each of $x = 16, 68/3, 34$ is attained at a unique point in the high strip. The relevant channel arithmetic is

$$\begin{array}{c|c|c} x & \text{channel} & \text{exact value of } \Delta \\ \hline 16 & (m;n,\ell) = (11;3,1) & 132 - 2 - 8 \cdot 16 = 2 > 0, \\ 68/3 & (10;2,6) & 110 - 42 - 3(68/3) = 0, \\ 34 & (11;2,5) & 132 - 30 - 3 \cdot 34 = 0. \end{array} \tag{9}$$

Thus the first split lies on the possible side. The latter two are exact channel walls, and their new modes are excluded by the strict convention.

The two localization consequences at k_{11} may also be read without the table:

$$x_{8/15} - 44 > \left(\frac{\pi_-}{1 - 8/15}\right)^2 - 44 = \frac{725209}{550564} > 0, \tag{10}$$

$$x_{1/4} - \frac{33}{2} > \left(\frac{\pi_-}{1 - 1/4}\right)^2 - \frac{33}{2} = \frac{5871}{5618} > 0. \tag{11}$$

Hence a second mode at k_{11} forces $\rho < 8/15$, and a third mode forces $\rho < 1/4$. Finally, the first-mode ball walls give

$$\left(\frac{69}{5}\right)^2 - 132 = \frac{1461}{25} = 44 + \frac{361}{25} = \frac{33}{2} + \frac{2097}{50}, \tag{12}$$

$$\left(\frac{64}{5}\right)^2 - 132 = \frac{796}{25} = \frac{33}{2} + \frac{767}{50}. \tag{13}$$

If $H = 10$ is present at k_{11}, then x is larger than both the second threshold 44 and the third threshold 33/2; hence every second and third mode is excluded. If $H = 9$ is present, every third mode is excluded. Equations (10)–(13) are the five formerly separate k_{11} localization implications.

5 The three impossible cap cells

If the first mode H is counted at k_m, its strict ball wall $\beta_{H,1} < k_m$ forces $x > \beta_{H,1}^2 - m(m+1)$. If the second mode h is counted, its product wall forces $x < (m(m+1) - h(h+1))/3$. The three dashed cells are therefore empty by the following exact, incompatible bounds:

$$\begin{array}{c|c|c|c} \text{cell} & \text{required lower bound} & \text{required upper bound} & \text{gap} \\ \hline k_8 : H = 6,\ h \geq 2 & 100 - 72 = 28 & (72 - 6)/3 = 22 & 6 \\ k_9 : H = 8,\ h \geq 0 & (71/6)^2 - 90 = 1801/36 & 90/3 = 30 & 721/36 \\ k_9 : H = 7,\ h \geq 3 & (23/2)^2 - 90 = 169/4 & (90 - 12)/3 = 26 & 65/4. \end{array} \tag{14}$$

No limiting equality can rescue a cell: both necessary modal inequalities are strict.

6 Checkpoint monotonicity and five global payments

Put $M = m(m+1)$ and $\mathcal{W}_m(\rho) := \mathcal{W}(\rho, k_m(\rho))$. Logarithmic differentiation shows that $\mathcal{W}_m'$ has the sign of

$$\pi^2(1 + \rho) - M\rho^2(1 - \rho)^2. \tag{15}$$

For $m \leq 11$,

$$\pi^2(1 + \rho) - M\rho^2(1 - \rho)^2 > 9 - \frac{132}{16} = \frac{3}{4} > 0.$$

Thus every $\mathcal{W}_m$, $m \le 11$, is strictly increasing. This single argument replaces both executable derivative-bound predicates.

At ρ_c, one has $z = \pi + 1/2 > 193/53$, $\rho_c < 1/7$, and $1/\pi > 7/22$. Consequently

$$\mathcal{W}(\rho_c, K) > \frac{2}{9}\frac{7}{22}\left(1 - \frac{1}{7^3}\right)K^3 = \frac{38}{539}K^3. \tag{16}$$

For each row in the next table, the middle column proves $k_m(\rho_c) > t_m$, and the last column proves $\mathcal{W}_m(\rho_c) > C_m$.

m	cap C_m	t_m	$(193/53)^2 + M - t_m^2$	$(38/539)t_m^3 - C_m$
7	40	208/25	67049/1755625	5083656/8421875
8	53	923/100	1901439/28090000	656778873/269500000
9	73	254/25	61431/1755625	7911557/8421875
10	89	111/10	14211/280900	1999489/269500
11	121	241/20	65271/1123600	5076899/2156000

Every last-column entry is positive. Monotonicity therefore extends each global payment from ρ_c to the complete high strip.

7 Seven conditional first-block payments

We now give a general reduction which avoids any numerical enclosure of a radical. Suppose the presence of first mode H below the checkpoint k_m forces $K > t$. Set

$$a^2 := t^2 - m(m+1), \qquad \rho_0 := 1 - \frac{\pi}{a}. \tag{17}$$

All seven rows below have $a > \pi$: the smallest listed radicand is

$$\frac{89}{4}, \qquad \frac{89}{4} - \left(\frac{22}{7}\right)^2 = \frac{2425}{196} > 0.$$

If the mode is present for some $K \le k_m(\rho)$, then $k_m(\rho) > t$, hence $x_\rho > a^2$ and $\rho > \rho_0$. By the strict increase of $\mathcal{W}_m$,

$$\mathcal{W}_m(\rho) > \mathcal{W}_m(\rho_0) = \mathcal{W}(\rho_0, t).$$

Choose a rational $r > \rho_0$. Since $\mathcal{W}(\rho, t)$ decreases with ρ at fixed t, Lemma 1 yields

$$\mathcal{W}(\rho_0, t) > \mathcal{W}(r, t) > \frac{7}{99}(1 - r^3)t^3. \tag{18}$$

The condition $r > \rho_0$ follows, without taking a square root, from

$$L := \pi_-^2 - (1 - r)^2 a^2 > 0. \tag{19}$$

Here is the complete calculation. The column P is the reserve in (18) above the stated cap.

m	H	t	cap	a^2	r	L	$P = \frac{7}{99}(1 - r^3)t^3 - \text{cap}$
7	6	10	53	44	8/15	725209/2528100	18659/2673
8	7	23/2	80	241/4	3/5	64349/280900	213281/49500

m	H	t	cap	a^2	r	L	P
9	8	71/6	81	1801/36	3/5	4713989/2528100	14506973/1336500
10	7	23/2	100	89/4	7/20	2105431/4494400	18539033/6336000
10	8	71/6	97	1081/36	3/7	317585/4955076	2865431/261954
10	9	64/5	100	1346/25	3/5	8811001/7022500	25143284/1546875
11	9	64/5	125	796/25	1/2	536261/280900	58757/12375

All L's and P's are positive. Each row therefore proves both the localization of its conditional branch and its Weyl payment.

8 The connected cap–74 path

The seventeenth localization row is tied to the exceptional $k_9, H = 6, h = 4$ cap as follows. Angular propagation from the strict $(3,2)$ Bessel wall gives

$$j_{9/2,2}^2 > \left(\frac{103}{10}\right)^2 + 20 - 12 = \frac{11409}{100} = 108 + \frac{609}{100}. \tag{20}$$

Accordingly the strict proxy uses $\beta_{4,2}^2 = 11409/100$; the strict inequality belongs to the actual Bessel zero $j_{9/2,2}$, not to the proxy constant. At k_9, the product wall for $(4,2)$ is

$$3x < 90 - 20 = 70, \qquad \text{i.e.} \qquad x < \frac{70}{3}.$$

The ball wall (20), on the other hand, forces

$$x > \frac{11409}{100} - 90 = \frac{2409}{100}, \qquad \frac{2409}{100} - \frac{70}{3} = \frac{227}{300} > 0.$$

Thus the fully sharpened cap–74 cell is empty.

For completeness, even the deliberately conservative cell is paid. At $\rho = 7/20$,

$$x_{7/20} - \frac{70}{3} > \frac{400}{169}\left(\frac{333}{106}\right)^2 - \frac{70}{3} = \frac{36230}{1424163} > 0, \tag{21}$$

so a counted $(4,2)$ mode forces $\rho < 7/20$. Also

$$108 - \left(\frac{207}{20}\right)^2 = \frac{351}{400} > 0, \tag{22}$$

so its ball wall forces $K > 207/20$. Therefore

$$\begin{aligned} \mathcal{W}(\rho, K) &> \frac{7}{99}\left(1 - \left(\frac{7}{20}\right)^3\right)\left(\frac{207}{20}\right)^3 \\ &= \frac{52823261673}{704000000} = 74 + \frac{727261673}{704000000} > 74. \end{aligned} \tag{23}$$

The cap itself is exactly $(6+1)^2 + (4+1)^2 = 49 + 25 = 74$. Equations (20)–(23) cover both the actual incompatibility and every row of the conservative cap–74 fallback.

9 Exact row map and scope audit

The following map uses the ordering of the two functions in the original exact registry. To keep the table readable, L denotes `verify_localizations`, and C denotes `verify_cap_tables_and_payments`. The map lets a referee match every replaced predicate to a displayed hand calculation.

registry family	row numbers	analytic replacement here
L	1–2	distinctness and connectedness: (6)–(7) and the seventeen-row table
L	3–19	all ratios in the high strip, by $\rho_c < 1/7 < 4/25$ and $2/3 < 7/8$
L	20–36	the seventeen exact sign witnesses in the long table
L	37–42	split location (8) and side/equality arithmetic (9)
L	43–47	the five k_{11} implications (10)–(13)
C	44–55	the twelve non-bookkeeping cap–74 predicates, (20)–(23); row $C56$, the block identity $49 + 25 = 74$, is allocated to Packet A
C	57–59	the three impossible cells, (14)
C	78–79	the general checkpoint monotonicity proof (15)
C	80–84	the five global-payment rows following (16)
C	85–98	the seven localization and seven conditional-payment rows in the final long table

The count reconciliation is

subfamily	connected mathematical objects	registry predicates
registry metadata	1	2
high-strip membership	17	17
localization signs	17	17
algebraic splits	3	6
special k_{11} implications	5	5
cap–74 path	1	12
impossible cells	3	3
checkpoint monotonicity	1	2
global payments	5	5
conditional payments	7	14
total	60	83.

In particular, the five headline Packet–C subfamilies contain exactly

$$17 + 3 + 3 + 5 + 7 = 35$$

connected objects (localization signs, algebraic splits, impossible cells, global payments, and conditional payments). Because each algebraic split has a location and a side predicate, and each conditional payment has a localization and a reserve predicate, these headline objects replace $17 + 6 + 3 + 5 + 14 = 45$ registry predicates. The remaining 38 predicates in the total are the explicitly listed auxiliary closure rows.

Thus this packet replaces all 47 predicates of L and 36 predicates of C. The elementary cap addition rows 1–43, row 56, and rows 60–77 of the latter family are outside the declared Packet C scope; they are instances of the general odd-sum/block-cap identity and are not claimed here. No row in the declared scope is unreconstructed.

Purely Analytic Ledger Replacement, Packet D: Seam Arithmetic and the Two-Sided Payments

Exact hand-checkable exhibit

Abstract

This packet replaces the twenty-two executable seam predicates and the eleven moving/fixed payment predicates used in the high side through k_6. Every premise is an analytic inequality already derived in the seam or staircase argument, and every remaining decision is displayed as an exact rational identity or strict rational comparison. In particular, the derivative of the seam comparison function is derived explicitly and is bounded by an exact proxy with a positive margin to 16/5. The six moving k_6 payments are printed with their full rational differences, not merely their numerators. No program execution, floating-point value, or interval calculation is a premise.

1 Rational constants used by both tables

For completeness, the needed bounds for π can be checked from the alternating arctangent series, rather than imported from an executable Machin enclosure. Machin's identity and the alternating remainder rule give

$$\pi > 16\left(\frac{1}{5} - \frac{1}{3\,5^3} + \frac{1}{5\,5^5} - \frac{1}{7\,5^7}\right) - \frac{4}{239} =: L_\pi,$$
$$\pi < 16\left(\frac{1}{5} - \frac{1}{3\,5^3} + \frac{1}{5\,5^5}\right) - 4\left(\frac{1}{239} - \frac{1}{3\,239^3}\right) =: U_\pi.$$

Indeed, the tangent-addition formula proves $\pi = 16\arctan(1/5) - 4\arctan(1/239)$, and the displayed truncations have the indicated directions. Direct rational subtraction gives

$$L_\pi - \frac{333}{106} = \frac{3418213}{41563593750} > 0, \qquad \frac{1571}{500} - U_\pi = \frac{323354809}{853244937500} > 0. \tag{1}$$

Since

$$\frac{22}{7} - \frac{1571}{500} = \frac{3}{3500} > 0, \tag{2}$$

we may use throughout

$$\underline{\pi} := \frac{333}{106} < \pi < \overline{\pi} := \frac{1571}{500} < \frac{22}{7}. \tag{3}$$

All later occurrences of a rational π bound refer to (3).

2 The twenty-two seam rows

2.1 Analytic input and comparison function

We recall precisely the analytic data to which the finite arithmetic is applied. Put

$$\epsilon = 1 - \rho = y^2, \qquad \kappa = K\epsilon^2, \qquad 0 < y \le \frac{1}{\sqrt{6}}, \quad \rho \ge \frac{5}{6}, \quad \kappa \ge 54.$$

For a first plateau of length p, and the quantities m, R in the seam decomposition, set

$$P = py, \qquad r = Ry, \qquad S = (p+m)y.$$

The angular estimate $d > 5/8$ and $R = p - dm$ imply, whenever $r > 0$,

$$0 < r \le P, \quad S \le S_*(P,r) := \frac{13P - 8r}{5}. \tag{4}$$

Equality in the second relation can occur when $m = 0$; no later step requires strictness at this comparison. The curvature-shelf calculation already proved in the seam argument gives the self-consistent inequality

$$P^2 < \frac{2\pi^2 Q}{(1-\widehat{q})^2}, \quad Q = 1 + \frac{y^2}{\kappa} + \frac{yS}{\kappa}, \quad \widehat{q} = \frac{y^2}{\rho}(Q-1). \tag{5}$$

Since the right-hand side increases with Q, define

$$Q_* = 1 + \frac{y^2}{\kappa} + \frac{yS_*}{\kappa}, \quad q_* = a(Q_* - 1), \qquad a := \frac{y^2}{\rho} \le \frac{1}{5}, \tag{6}$$

and

$$H(P,r) := \frac{2\pi^2 Q_*}{(1-q_*)^2}. \tag{7}$$

Then (4) gives $Q \le Q_*$ and $\widehat{q} \le q_*$. Since (5) is itself strict and its right-hand side increases with Q, it follows that $P^2 < H(P,r)$.

The remaining arithmetic is exactly the following table. Rows 1–5 are the five constant/shelf rows, rows 6–10 are the five displacement rows, rows 11–14 are the four derivative rows, rows 15–17 are the three endpoint rows, and rows 18–22 are the five terminal-payment rows. Thus the table reconciles one-for-one with all twenty-two seam predicates.

For clarity, the radical used in rows 7–9 follows directly from the analytic shelf bound. Namely,

$$p^2 < \frac{2\pi\rho K}{\epsilon}, \quad P = py, \quad K = \frac{\kappa}{y^4}$$

imply

$$\left(\frac{y^3 P}{\kappa\rho}\right)^2 < \frac{2\pi y^2}{\kappa\rho} \le \frac{\pi}{135} < \frac{\overline{\pi}}{135}.$$

Together with $S_* \le 13P/5$, this gives

$$q_* < \frac{1}{1620} + \frac{13}{5}\frac{763}{5000},$$

which is precisely the dependency recorded by rows 6–9.

#	quantity	exact rational check	role
1	seam π bound	$\overline{\pi} - U_\pi = 323354809/853244937500 > 0$	$\pi < \overline{\pi}$; (1)
2	lower $\sqrt{2}$ bound	$2 - (140/99)^2 = 2/9801 > 0$	$140/99 < \sqrt{2}$
3	upper $\sqrt{2}$ bound	$(99/70)^2 - 2 = 1/4900 > 0$	$\sqrt{2} < 99/70$
4	y ceiling	$(49/120)^2 - 1/6 = 1/14400 > 0$	$y < 49/120$
5	shelf ceiling	$(13/5)^2 - 2\overline{\pi} = 119/250 > 0$	$\sqrt{2\pi\rho} < 13/5$; rows 1

#	quantity	exact rational check	role
6	first displacement term	$\frac{(1/6)^2}{54(5/6)} = \frac{1}{1620}$	$y^4/(\kappa\rho) \leq 1/1620$
7	radical ceiling	$(763/5000)^2 - \frac{\overline{\pi}}{135} = \frac{8563}{675000000} > 0$	shelf radical; row 1
8	displacement proxy	$\frac{1}{1620} + \frac{13}{5}\frac{763}{5000} = \frac{804689}{2025000}$	rows 6–7
9	displacement margin	$\frac{159}{400} - \frac{804689}{2025000} = \frac{497}{4050000} > 0$	$q_*, \widehat{q} < \overline{q} := 159/400$
10	outer-half localization	$\frac{5}{6}(1 - 159/400) = \frac{241}{480} = \frac{1}{2} + \frac{1}{480}$	$x_0/K > 1/2$; row 9
11	Q_*-derivative proxy	$D_Q := \frac{13}{5}\frac{49/120}{54} = \frac{637}{32400}$	$\partial_P Q_* < D_Q$; row 4
12	exact H-derivative proxy	$\frac{2\overline{\pi}^2 D_Q(1 + 159/400 + 2/5)}{(1 - 159/400)^3} = \frac{2260740364246}{708624500625}$	rows 1, 9, 11 and $a \leq 1/5$
13	margin to 16/5	$\frac{16}{5} - \frac{2260740364246}{708624500625} = \frac{6858037754}{708624500625} > 0$	$\partial_P H(P, B) < 16/5$
14	synthetic-slope margin	$2\frac{14}{3} - \frac{16}{5} = \frac{92}{15} > 0$	$\partial_P(P^2 - H) > 92/15$
15	endpoint Q_* proxy	$\overline{Q} := 1 + \frac{1/6}{54} + \frac{(49/120)(14/3)}{54} = \frac{10093}{9720}, \quad \overline{Q} - 1 = \frac{373}{9720}$	$Q_*(B, B) < \overline{Q}$
16	endpoint q_* proxy	$\frac{1}{5}(\overline{Q} - 1) = \frac{373}{48600}$	$q_*(B, B) < 373/48600$
17	initial reserve	$\left(\frac{14}{3}\right)^2 - \frac{2\overline{\pi}^2(10093/9720)}{(1 - 373/48600)^2} = \frac{2505132463469}{2616573970125} > 0$	$B^2 - H(B, B) > 0$; rows 1, 15–16
18	action-gain coefficient	$g := \frac{2(140/99)54}{3(1571/500)} = \frac{280000}{17281}$	$K\eta > g/y$; rows 1–2
19	gain after R loss	$g - \frac{14}{3} = \frac{598066}{51843}$	use $R < 14/(3y)$
20	interface loss proxy	$\frac{8}{15}\frac{99}{70} = \frac{132}{175}$	$4\delta < 132/175$; row 3
21	dangerous-branch reserve	$\frac{598066}{51843}\frac{120}{49} - 1 - \frac{132}{175} = \frac{80132733}{3024175} > 0$	$R > 0$; rows 4, 19–20
22	safe-branch reserve	$\frac{280000}{17281}\frac{120}{49} - 1 - \frac{132}{175} = \frac{114694733}{3024175} > 0$	$R \leq 0$; rows 4, 18, 20

2.2 Why the table proves the seam implications

We spell out the two implications for which the table is more than a collection of arithmetic facts.

First, on the segment $r = B := 14/3$ and $P \geq B$,

$$\partial_P Q_* = \frac{13y}{5\kappa}, \quad \partial_P q_* = a\,\partial_P Q_*.$$

Differentiation of (7), followed only by the identity $aQ_* = q_* + a$, gives

$$\partial_P H = 2\pi^2(\partial_P Q_*)\frac{1 - q_* + 2aQ_*}{(1 - q_*)^3} = 2\pi^2(\partial_P Q_*)\frac{1 + q_* + 2a}{(1 - q_*)^3}. \tag{8}$$

Rows 1, 9, and 11 and $a \le 1/5$ substitute monotonically into (8); rows 12–13 then prove $\partial_P H < 16/5$. Because $P \ge B$, row 14 proves

$$\frac{d}{dP}\big(P^2 - H(P,B)\big) = 2P - \partial_P H(P,B) > \frac{92}{15} > 0. \tag{9}$$

Rows 15–17 prove $B^2 - H(B,B) > 0$. Integration of (9) therefore yields $P^2 > H(P,B)$ for every $P \ge B$. If $r \ge B$, then $H(P,r) \le H(P,B)$ because $S_*(P,r)$ decreases with r and H increases with Q_*. This contradicts $P^2 < H(P,r)$. Hence

$$r < B, \quad R < \frac{14}{3y}. \tag{10}$$

Second, the turning-point action bound gives

$$K\eta \ge \frac{2\sqrt{2}\,\kappa}{3\pi}\frac{1}{y} > \frac{g}{y},$$

where row 18 is the exact rational lower proxy. Also row 4 gives $1/y > 120/49$, and row 20 gives $4\delta < 8\sqrt{2}/15 < 132/175$. With $M = \lfloor K\eta \rfloor - R$ and the strict inequality $\lfloor x \rfloor > x - 1$, (10) and row 21 give, when $R > 0$,

$$M - 4\delta > \left(g - \frac{14}{3}\right)\frac{120}{49} - 1 - \frac{132}{175} > 0.$$

When $R \le 0$, row 22 gives the still stronger estimate

$$M - 4\delta > g\frac{120}{49} - 1 - \frac{132}{175} > 0.$$

Thus the seam tail has the strict action payment claimed in the main argument, including the $R = 0$ and integer-$K\eta$ faces.

Remark 1 (Notation correction in the executable label). The supplemental verifier labels row 11 as an "exact dq/dP cap." In the notation of the proof, the number 637/32400 is a proxy for $\partial_P Q_*$, not for $\partial_P q_*$. In fact $\partial_P q_* = a\partial_P Q_*$ and hence $\partial_P q_* < 637/162000$. Formula (8) and the verifier both use 637/32400 in its correct role as the Q_*-derivative proxy. The proof is therefore sound, but the old ledger label is inaccurate.

Remark 2 (Source wording for the π bound). One sentence in the seam passage groups $\pi < 1571/500$ with bounds said to be proved by positive rational squaring. Squaring proves rows 2–5, but not a bound for π. Equations (1)– (3) supply the required analytic alternating-series proof.

3 The eleven k_6 payment rows

Put

$$z_\rho = \frac{\pi}{1-\rho}, \quad k_m(\rho)^2 = z_\rho^2 + m(m+1), \quad p_1(\rho)^2 = 4z_\rho^2 + 2, \tag{11}$$

and

$$\mathcal{W}(\rho, K) = \frac{2}{9\pi}(1 - \rho^3)K^3. \tag{12}$$

For a moving wall $T^2 = az_\rho^2 + b$, direct differentiation shows that $\mathcal{W}(\rho, T)$ increases whenever

$$a\pi^2(1 + \rho) > b\rho^2(1 - \rho)^2. \tag{13}$$

Here $b/a \le 42$. Since $\rho^2(1-\rho)^2 \le 1/16$ and $\pi > 3$, the left side of (13), after division by a, is greater than 9, whereas the right side is at most $42/16$; hence every moving payment in the table is increasing. At a fixed wall L, the function $\mathcal{W}(\rho, L)$ is decreasing in ρ.

For a rational split r, define

$$A(a,b;r) := a\frac{(333/106)^2}{(1-r)^2} + b, \quad B(r) := \frac{7}{99}(1-r^3). \tag{14}$$

The bounds (3) give, on a moving wall at $\rho = r$,

$$\mathcal{W}(r,T) > B(r)A(a,b;r)^{3/2}. \tag{15}$$

Thus, for a positive cap C, the rational inequality

$$\Delta_{\text{mov}} := B(r)^2 A(a,b;r)^3 - C^2 > 0 \tag{16}$$

implies the strict payment, because

$$\Delta_{\text{mov}} = \big(B(r)A^{3/2} - C\big)\big(B(r)A^{3/2} + C\big)$$

and the second factor is positive. At a fixed wall L, it is enough to check the ordinary rational reserve

$$\Delta_{\text{fix}} := B(r)L^3 - C > 0. \tag{17}$$

The following is the complete eleven-row registry: six moving rows and five fixed rows. The fractions in the last column are the full reduced differences in (16) or (17).

wall/branch	split r	wall data	cap C	full exact rational difference
k_1 moving	1/8	$(a,b) = (1,2)$	4	$\Delta_{\text{mov}} = \dfrac{153864824754340538785}{348796101192617852928} > 0$
k_2 moving	3/10	$(a,b) = (1,6)$	9	$\Delta_{\text{mov}} = \dfrac{1399810848073457853053}{3942374914015230000000} > 0$
k_3 moving	1/2	$(a,b) = (1,12)$	16	$\Delta_{\text{mov}} = \dfrac{137027505353736631}{514922437748928} > 0$
k_4 moving	1/2	$(a,b) = (1,20)$	25	$\Delta_{\text{mov}} = \dfrac{2507119282010251109}{13902905819221056} > 0$
k_5 moving	13/25	$(a,b) = (1,30)$	36	$\Delta_{\text{mov}} = \dfrac{92672404686738742434741}{709259556128000000000} > 0$
p_1 moving	1/5	$(a,b) = (4,2)$	29	$\Delta_{\text{mov}} = \dfrac{373255772037478993002833}{868931613701316000000} > 0$
k_2 fixed	3/10	$L = 51/10$	9	$\Delta_{\text{fix}} = \dfrac{1387329}{11000000} > 0$
k_3 fixed	1/2	$L = 13/2$	16	$\Delta_{\text{fix}} = \dfrac{6277}{6336} > 0$
k_4 fixed	1/2	$L = 15/2$	25	$\Delta_{\text{fix}} = \dfrac{775}{704} > 0$
k_5 fixed	13/25	$L = 17/2$	36	$\Delta_{\text{fix}} = \dfrac{452843}{343750} > 0$
p_1 fixed	1/5	$L = 77/10$	29	$\Delta_{\text{fix}} = \dfrac{849901}{281250} > 0$

We now verify that the table pays every branch rather than merely listing positive numbers. For each maximum entry wall

$$\max\{k_2, 51/10\}, \quad \max\{k_3, 13/2\}, \quad \max\{k_4, 15/2\}, \quad \max\{k_5, 17/2\}, \quad \max\{p_1, 77/10\},$$

partition the ratio interval at the displayed r. On $\rho \ge r$, the maximum is at least the moving component and monotonicity together with the moving row pays C. On $\rho \le r$, the maximum is at least the fixed component and the fixed row pays C. This argument does not assume that the two components are equal at the rational split. For k_1, the applicable ratio interval starts at $\rho_c > 1/8$ (because $\pi < 22/7 < 7/2$), so the first moving row applies directly. Strict payments also cover coincident entry walls, while the wall itself remains assigned to the lower cap by the strict spectral convention.

4 Reconciliation with the lower-d rows

The two rational π predicates used by the two-sided ledger are already proved in (3). The eleven k_6 predicates are exactly the eleven rows in the preceding table. The remaining thirteen wall/cap predicates in that ledger are the ten transparent multiplicity sums

$$\begin{aligned}(1,\ 1+3,\ 1+3+5,\ 1+3+5+1,\ 1+3+5+7,\\ 1+3+5+7+1,\ 1+3+5+7+9+1,\\ 1+3+5+7+9+1+3,\\ 1+3+5+7+9+11+1+3,\\ 1+3+5+7+9+11+1+3+5)\\ =(1,4,9,10,16,17,26,29,40,45),\end{aligned} \tag{18}$$

and the endpoint ordering $13/2 < 2z_\rho < 15/2$. The latter follows without an executable check: using $\sqrt{337} < 19$ on the lower endpoint and (3),

$$2\frac{333}{106}\frac{19}{18} - \frac{13}{2} = \frac{7}{53} > 0, \qquad \frac{15}{2} - \left(2\frac{22}{7}+1\right) = \frac{3}{14} > 0. \tag{19}$$

Thus the cap–10 row is empty, and the three executable ordering rows collapse to (19) and its immediate empty-cell consequence.

The nine lower-d payment rows need no new table here: the main paper's lemma "Lower side through d" (label `cm:lem-lower-d`) already prints the seven fixed-wall differences

$$\frac{793292}{1546875}, \frac{486236661}{1375000000}, \frac{333100003}{99000000}, \frac{329797}{88000}, \frac{3590476297}{1125000000}, \frac{327078887}{99000000}, \frac{8805519}{1375000},$$

and states the initial and moving payments $\mathcal{W}(\rho_c, L(\rho_c)) > 19/6$, where $L(\rho) = (2\rho)^{-1}$ on this branch, and $\mathcal{W}(1/19, 2z) > 508/27$. If desired, their exact printed lower proxies close by the two one-line subtractions

$$\frac{57421}{16854} - \frac{19}{6} = \frac{675}{2809} > 0, \qquad \frac{173863}{8427} - \frac{508}{27} = \frac{137795}{75843} > 0. \tag{20}$$

Consequently every seam and two-sided payment predicate assigned to Packet D is now visible as analytic reasoning plus printed integer arithmetic.

5 Dependency count and conclusion

The replacement count is exact:

$$\underbrace{22}_{\text{seam table}} + \underbrace{(2+6+5+10+3)}_{k_6/\text{lower-}d\text{ wall–cap ledger}} + \underbrace{9}_{\text{already printed lower-}d\text{ payments}}.$$

The two constant rows are supplied by (3); the six plus five payment rows are the eleven-row table; the ten bookkeeping rows are (18); the three ordering rows are (19); and the nine already printed payments are explicitly cross-referenced above. This removes program execution from the logical dependency of precisely the Packet-D portion of the exact ledger.

Upper-Owner and Optical Cleanup

Analytic ledger replacement, Packet E

Abstract

This note replaces the executable scope checks surrounding the upper owner of the final spherical-shell residual. It proves directly that the eligible upper wall is $K_0(\rho)$ and that $k_{11}(\rho) < K_0(\rho)$ on the required ratio interval. It also records the symbolic identities that make the repeated optical ledger samples unnecessary.

1 The upper owner

For $0 < \rho < 1$, set

$$a_\rho := \frac{2\pi\rho}{1-\rho}, \qquad \eta_\rho := G_1(\max\{\rho, 1/2\}),$$

where

$$G_1(s) = \frac{\sqrt{1-s^2} - s\arccos s}{\pi}, \qquad \omega_0 := G_1(1/2) = \frac{\sqrt{3}}{2\pi} - \frac{1}{6}.$$

Let $C_0 > 0$, and define

$$K_0(\rho) := \left(\frac{\sqrt{a_\rho} + \sqrt{a_\rho + 4\eta_\rho C_0}}{2\eta_\rho} \right)^2. \tag{1}$$

Also put

$$\rho_c := \frac{1}{1+2\pi}, \qquad z_\rho := \frac{\pi}{1-\rho}, \qquad k_{11}(\rho) := \sqrt{z_\rho^2 + 132}.$$

Lemma 1 (The final upper wall is genuinely above the staircase)**.** *For*

$$\rho_c \le \rho < \frac{7}{8},$$

one has

$$k_{11}(\rho) < 64 < U(\rho). \tag{2}$$

More precisely, $U(\rho) = K_0(\rho) > 64$ *for* $\rho_c \le \rho < 5/6$*, while*

$$U(\rho) = \frac{54}{(1-\rho)^2} \ge 1944 \qquad (5/6 \le \rho < 7/8).$$

In particular, on the final surviving interval $\rho_c \le \rho < 39/50$*, the upper owner is* $K_0(\rho)$*.*

Proof. We use only

$$3 < \pi < \frac{22}{7}, \qquad \sqrt{3} < \frac{7}{4}.$$

They imply

$$\omega_0 = \frac{\sqrt{3}}{2\pi} - \frac{1}{6} < \frac{7/4}{6} - \frac{1}{6} = \frac{1}{8}.$$

Since

$$G_1'(s) = -\frac{\arccos s}{\pi} < 0 \quad (0 \le s < 1), \qquad G_1(1) = 0,$$

one has $0 < \eta_\rho \le \omega_0 < 1/8$. The function a_ρ is increasing, and its value at ρ_c is exactly one:

$$a_{\rho_c} = \frac{2\pi/(1+2\pi)}{2\pi/(1+2\pi)} = 1.$$

Thus $a_\rho \ge 1$. Formula (1) and $C_0 > 0$ give

$$\sqrt{K_0(\rho)} > \frac{\sqrt{a_\rho}}{\eta_\rho} > 8,$$

so $K_0(\rho) > 64$.

On the other hand, $\rho < 7/8$, and therefore

$$z_\rho < 8\pi < \frac{176}{7} < 26.$$

Consequently

$$k_{11}(\rho)^2 < 26^2 + 132 = 808 < 64^2,$$

which proves (2).

The small-hole wall H_0 is eligible only when $\rho < \omega_0$. But

$$\rho_c > \frac{1}{8} > \omega_0,$$

because $\pi < 22/7 < 7/2$ implies $1 + 2\pi < 8$. Also $\pi > 3$ gives the exact scope chain

$$\rho_c < \frac{1}{7} < \frac{39}{50} < \frac{5}{6}.$$

For $\rho < 5/6$, the thin-seam wall is not eligible either, so the piecewise upper-owner formula gives $U = K_0 > 64$. For $5/6 \le \rho < 7/8$, the seam branch owns the equality face and

$$U(\rho) = \frac{54}{(1-\rho)^2} \ge \frac{54}{(1/6)^2} = 1944 > 64.$$

Combining this with $k_{11} < 64$ proves (2). Finally $39/50 < 5/6$, so the final surviving interval lies wholly in the K_0 branch. $\square$

2 Symbolic optical identities

The optical ledger evaluates the same algebraic statements at several sample values. The following identities replace those repetitions.

For $a/\pi = N + t$, with $N \ge 1$ and $0 < t \le 1$, the exact continuous-minus-product deficit is

$$\frac{D(a)}{\pi^2} = \frac{N^2}{2} + N\left(t^2 + \frac{1}{6}\right) + \frac{2t^3}{3}. \tag{3}$$

If $C_D = 1382/3125$, direct subtraction gives

$$\frac{D(a)}{\pi^2} - C_D(N+t)^2 = \left(\frac{1}{2} - C_D\right) N^2 + N\left(t^2 - 2C_D t + \frac{1}{6}\right) + \frac{2t^3}{3} - C_D t^2. \tag{4}$$

The coefficient of N^2 is positive. The difference between the right sides at $N+1$ and N is

$$2\left(\frac{1}{2} - C_D\right) N + \left(\frac{1}{2} - C_D\right) + t^2 - 2C_D t + \frac{1}{6}.$$

The quadratic $t^2 - 2C_D t + 1/6$ has minimum $1/6 - C_D^2$ at $t = C_D$. Consequently, for $N \geq 1$, the displayed difference is at least

$$3\left(\frac{1}{2} - C_D\right) + \frac{1}{6} - C_D^2 = \frac{4229603}{29296875} > 0.$$

Therefore the deficit increases in N, and it is enough to check $N = 1$. At $N = 1$, differentiation gives

$$\frac{d}{dt}\left(\frac{D(a)}{\pi^2} - C_D(1+t)^2\right) = 2(t - C_D)(t+1).$$

Thus the unique minimum on $0 < t \leq 1$ occurs at $t = C_D$, where the exact value is

$$\frac{1822532}{91552734375} > 0. \tag{5}$$

This proves the product screen simultaneously for every N, t.

Likewise, the twelve repeated angular block checks are all instances of

$$\sum_{\ell=0}^{m-1} (2\ell + 1) = m^2 \qquad (m \in \mathbb{N}_0), \tag{6}$$

proved by subtracting consecutive squares. At a radial equality wall the strict convention excludes the newly proposed layer, so no boundary sample is added. These symbolic statements cover every repeated optical row; the two exact endpoint reserves remain those printed in the main optical lemma.

3 The two optical screens with all scaling shown

For completeness, we now print the algebra that was compressed in the main optical lemma. Set

$$\varepsilon_0 = \frac{11}{50}, \qquad c = \frac{1126}{625}, \qquad q = \frac{106}{333}, \qquad C_D = \frac{1382}{3125}.$$

The inequalities

$$\pi > \frac{333}{106}, \qquad \pi < \frac{22}{7}$$

give $q > 1/\pi$, and $0 < \varepsilon \leq \varepsilon_0 < 1/2$.

Lemma 2 (Complete optical screen arithmetic)**.** *Let $a = \varepsilon K$.*

1. *If $\pi < a \leq c/\varepsilon$, then*

$$C_D > \frac{2cq}{3} + \frac{\varepsilon}{4} + \varepsilon^2 q^2. \tag{L}$$

2. If $a \geq c/\varepsilon$, define

$$L(\varepsilon) := \frac{(1-\varepsilon)^2}{4} - \frac{(22/7)(1-\varepsilon)\varepsilon}{4c},$$
$$A(\varepsilon) := q\varepsilon + \frac{\varepsilon^2}{4c} + \frac{q\varepsilon^3}{4c} + \frac{\varepsilon^4}{16c^2}.$$

Then $L(\varepsilon) > A(\varepsilon)$.

3. The low and high domains are both closed at $a = c/\varepsilon$, so their union covers every $a > \pi$.

Proof. For the low screen, $D(a) > C_D a^2$ was proved above. The sufficient product inequality obtained in the main optical argument is

$$D(a) \geq \varepsilon\left(\frac{2a^3}{3\pi} + \frac{a^2}{4}\right) + \frac{\varepsilon^2 a}{\pi}. \tag{1}$$

After division by a^2, the three terms on the right are bounded by

$$\frac{2\varepsilon a}{3\pi} \leq \frac{2cq}{3}, \qquad \frac{\varepsilon}{4} \leq \frac{\varepsilon_0}{4}, \qquad \frac{\varepsilon^2}{\pi a} < \frac{\varepsilon^2}{\pi^2} < \varepsilon_0^2 q^2.$$

Exact subtraction gives

$$C_D - \left(\frac{2cq}{3} + \frac{11}{200} + \left(\frac{11}{50}\right)^2 q^2\right) = \frac{39569}{2772225000} > 0. \tag{7}$$

This proves (L). For $0 \leq a \leq \pi$, the strict radial count is zero, including $a = \pi$, so no low-frequency gap remains.

For the high screen, the exact radial deficit and angular ceiling error from the main analytic argument are

$$\frac{D_{\mathrm{rad}}}{a^2} \geq \frac{(1-\varepsilon)^2}{4} - \frac{\pi(1-\varepsilon)}{4a},$$
$$\frac{E}{a^2} = \frac{\varepsilon}{\pi} + \frac{\varepsilon^2}{4\pi a} + \frac{\varepsilon}{4a} + \frac{\varepsilon^2}{16a^2}.$$

Since $a \geq c/\varepsilon$, one has $1/a \leq \varepsilon/c$. Using the two rational bounds for π therefore gives

$$\frac{D_{\mathrm{rad}}}{a^2} \geq L(\varepsilon), \qquad \frac{E}{a^2} < A(\varepsilon).$$

Moreover,

$$L'(\varepsilon) = -\frac{1-\varepsilon}{2} - \frac{22/7}{4c}(1-2\varepsilon) < 0 \quad (0 < \varepsilon < 1/2),$$

whereas $A'(\varepsilon) > 0$ coefficientwise. Hence

$$L(\varepsilon) - A(\varepsilon) \geq L(\varepsilon_0) - A(\varepsilon_0).$$

The endpoint subtraction is

$$\begin{aligned} L(\varepsilon_0) - A(\varepsilon_0) &= \frac{1}{4}\left(\frac{39}{50}\right)^2 - \frac{22}{7}\frac{(39/50)(11/50)}{4(1126/625)} \\ &\quad - \left[\frac{11}{50}q + \frac{(11/50)^2}{4(1126/625)} + \frac{(11/50)^3 q}{4(1126/625)} + \frac{(11/50)^4}{16(1126/625)^2}\right] \\ &= \frac{14817541}{472867032960000} > 0. \end{aligned} \tag{8}$$

Thus the radial deficit strictly exceeds the angular error.

Finally,

$$\frac{c}{\varepsilon} \geq \frac{c}{\varepsilon_0} = \frac{2252}{275} > \frac{22}{7} > \pi, \qquad \frac{2252}{275} - \frac{22}{7} = \frac{9714}{1925} > 0.$$

Therefore the common face lies in the low domain as well as the high domain, and

$$\{a : \pi < a \leq c/\varepsilon\} \cup \{a : a \geq c/\varepsilon\} = \{a : a > \pi\},$$

and equality $a = c/\varepsilon$ belongs to both pieces. Together with the zero-count interval $0 \leq a \leq \pi$, this covers every $a \geq 0$. At $K = a = 0$, both the spectral count and the Weyl term vanish. $\square$

Ledger rows replaced. Lemma 1 replaces the eight executable `verify_u_and_k11` scope checks. Equations (3)–(6) replace the repeated product samples and odd-block identities. These are rows of `verify_optical_constants`. Lemma 2 prints the remaining scaling, monotonicity, common-face, and exact-reserve implications. Thus all fifty optical rows are consequences of displayed identities rather than executable predicates.

Exact Owner Subtraction and Direct Analytic Closure

Analytic ledger replacement, Packet F

Abstract

The executable exact ledger checks a 9×7 truth table when the accepted lower, staircase, and optical regions are subtracted from the Round–19 residual. This note replaces all sixty-three state checks and the five additional interface checks by one set-theoretic calculation. The surviving set is then placed directly inside the closed domain of the all-frequency retained-remainder theorem. Every open and closed face is retained explicitly, and no artificial frequency interface is introduced.

1 Sets and inherited owners

Let

$$0 < \rho_* < \rho_0 < \rho_c < \rho_{\rm opt} < \frac{7}{8}, \qquad \rho_{\rm opt} := \frac{39}{50},$$

and let $U(\rho)$, $k_6(\rho)$, and $k_{11}(\rho)$ be the upper owner and the two staircase walls from the main proof. Put

$$\begin{aligned} D_{\rm low} &:= \left\{\rho_* < \rho \le \rho_0,\ (2\rho)^{-1} < K < U(\rho)\right\} \\ &\quad \cup \left\{\rho_0 < \rho < \rho_c,\ d < K < U(\rho)\right\}, \end{aligned} \tag{1}$$

$$D_{\rm high} := \left\{\rho_c \le \rho < \frac{7}{8},\ k_6(\rho) < K < U(\rho)\right\}, \tag{2}$$

where $d = (2\rho_0)^{-1}$, and define

$$D_{19} := D_{\rm low} \,\dot\cup\, D_{\rm high}. \tag{3}$$

The dot is justified by the disjoint ratio ranges.

The already-proved inclusive owners used in the subtraction are:

$$L_{\rm own} := D_{\rm low}, \tag{4}$$

$$S_{\rm own} := \left\{\rho_c \le \rho < \frac{7}{8},\ k_6(\rho) < K \le k_{11}(\rho)\right\}, \tag{5}$$

$$O_{\rm own} := \left\{\rho_{\rm opt} \le \rho < \frac{7}{8},\ K \ge 0\right\}. \tag{6}$$

The lower owner includes the complete $\rho = \rho_0$ fibre. The staircase includes its upper wall $K = k_{11}(\rho)$. The optical owner includes its lower ratio face $\rho = \rho_{\rm opt}$. In contrast, D_{19} already excludes $K = U(\rho)$.

2 The subtraction

Proposition 1 (Exact residual identity)**.** *One has*

$$\boxed{D_{19} \setminus (L_{\mathrm{own}} \cup S_{\mathrm{own}} \cup O_{\mathrm{own}}) = D_{20},} \tag{7}$$

where

$$D_{20} := \{\rho_c \le \rho < \rho_{\mathrm{opt}},\ k_{11}(\rho) < K < U(\rho)\}\,. \tag{8}$$

Proof. Since $L_{\mathrm{own}} = D_{\mathrm{low}}$ and the union in (3) is disjoint,

$$D_{19} \setminus L_{\mathrm{own}} = D_{\mathrm{high}}. \tag{9}$$

Inside D_{high}, the lower frequency inequality is strict: $k_6(\rho) < K$. Removing the inclusive staircase (5) therefore gives exactly

$$D_{\mathrm{high}} \setminus S_{\mathrm{own}} = \left\{\rho_c \le \rho < \frac{7}{8},\ k_{11}(\rho) < K < U(\rho)\right\}. \tag{10}$$

There is no missing equality case: $K = k_{11}(\rho)$ belongs to S_{own}, while $K = U(\rho)$ never belonged to D_{19}.

Finally, subtracting the inclusive ratio owner (6) changes $\rho < 7/8$ in (10) to $\rho < \rho_{\mathrm{opt}}$. Its equality face is removed because $\rho = \rho_{\mathrm{opt}}$ belongs to O_{own}. The face $\rho = \rho_c$ remains, since it belongs neither to D_{low} nor to O_{own}. The result is precisely (8). □

3 The five delicate interfaces

For clarity, the five extra assertions formerly checked separately by the executable ledger are consequences of the displayed sets:

1. At $\rho = \rho_0$, the identity $(2\rho_0)^{-1} = d$ makes the prospective second lower fibre empty; the first set in (1) owns the full equality fibre, and L_{own} removes it exactly once.
2. The face $\rho = \rho_c$ belongs to D_{high}. It is not an optical face, so points with $k_{11}(\rho_c) < K < U(\rho_c)$ remain in D_{20}.
3. The face $\rho = \rho_{\mathrm{opt}} = 39/50$ belongs to O_{own} and is therefore absent from D_{20}.
4. The frequency face $K = k_{11}(\rho)$ belongs to S_{own}, whereas $K = U(\rho)$ is excluded already by the strict inequality in D_{19}.
5. If $\rho_c < \rho < \rho_{\mathrm{opt}}$ and $k_{11}(\rho) < K < U(\rho)$, the point is in D_{high} but in none of $L_{\mathrm{own}}, S_{\mathrm{own}}, O_{\mathrm{own}}$; hence it survives, and every surviving point has exactly these properties.

4 Direct retained-remainder ownership

Let

$$\mathcal{R}_{\mathrm{rr}} := \{(\rho, K) : \rho_c \le \rho \le \rho_{\mathrm{opt}}, \quad K \ge k_{11}(\rho)\}\,. \tag{11}$$

This is the closed domain of the all-frequency retained-remainder theorem proved by the structural, angular-block, and scalar-positivity modules. In the notation of the main proof, that theorem states

$$(\rho, K) \in \mathcal{R}_{\mathrm{rr}} \quad \Longrightarrow \quad N_D(A_{\rho,1}, K^2) < W(\rho, K). \tag{12}$$

Proposition 2 (Direct ownership of the exact residual)**.** *One has*

$$\boxed{D_{20} \subset \mathcal{R}_{\mathrm{rr}}}, \qquad D_{20} \setminus \mathcal{R}_{\mathrm{rr}} = \varnothing. \tag{13}$$

Consequently the strict Pólya inequality holds at every point of D_{20}.

Proof. If $(\rho, K) \in D_{20}$, then

$$\rho_c \le \rho < \rho_{\mathrm{opt}} \quad \text{and} \quad k_{11}(\rho) < K < U(\rho).$$

The first pair of inequalities implies $\rho_c \le \rho \le \rho_{\mathrm{opt}}$, and the second implies $K \ge k_{11}(\rho)$. Hence $(\rho, K) \in \mathcal{R}_{\mathrm{rr}}$, which proves (13); the final assertion follows from (12).

The inclusion does not change any inherited owner. The face $\rho = \rho_{\mathrm{opt}}$ and the frequency face $K = k_{11}(\rho)$ are outside D_{20}, although the stronger theorem also proves them. The face $K = U(\rho)$ remains excluded from D_{20}, while $\rho = \rho_c$ remains included subject to the two strict frequency inequalities. There is no additional frequency face. □

Ledger rows replaced. Proposition 1 replaces all $9 \cdot 7 = 63$ state evaluations in the executable family `verify_exact_subtraction`. The five items above replace its five separately asserted interface predicates. Proposition 2 supplies the final residual closure by one set inclusion, without a frequency classifier or executable truth table.

A Global Structural Reduction for the Spherical-Shell Proxy

Global analytic structural module

Abstract

This note gives the global structural component of the purely analytic middle-ratio argument for the spherical-shell proxy. The weighted proxy is first written as an exact layer-cake sum. Its deficit from the action integral is then decomposed into a radial shifted-quadrature deficit and an angular rounding defect. A Stieltjes calculation retains two nonnegative remainders that are discarded by the optical estimate, and one of those remainders admits a smooth explicit lower bound. The result is the concrete sufficient inequality (37). All structural identities hold for every $0 < \rho < 1$ and $K > 0$; in particular they apply on the upward-closed middle-ratio region $\rho_c \le \rho \le 39/50,\ K \ge k_{11}(\rho)$.

1 Action and the global target

For $a > 0$ and $z \ge 0$, put

$$G_a(z) = \begin{cases} \dfrac{1}{\pi}\left(\sqrt{a^2 - z^2} - z \arccos \dfrac{z}{a}\right), & 0 \le z < a, \\ 0, & z \ge a. \end{cases} \tag{1}$$

Fix $0 < \rho < 1$ and $K > 0$, and define

$$A(z) = A_{\rho,K}(z) := G_K(z) - G_{\rho K}(z), \qquad T := A(0) = \frac{(1-\rho)K}{\pi}. \tag{2}$$

The strict positive-integer count is

$$[x]_< := \#\{n \in \mathbb{N} : n < x\}.$$

The spherical-shell proxy and its continuous payment are

$$P(\rho, K) := \sum_{\substack{\nu=\ell+1/2<K \\ \ell\in\mathbb{N}_0}} 2\nu \left[A(\nu) + \tfrac{1}{4}\right]_<, \tag{3}$$

$$W(\rho, K) := \int_0^K 2zA(z)\,dz = \frac{2(1-\rho^3)K^3}{9\pi}. \tag{4}$$

The upward-closed middle-ratio region used below is

$$\mathcal{R}_\mathrm{c} := \left\{(\rho, K) : \rho_c \le \rho \le \frac{39}{50}, \quad K \ge k_{11}(\rho)\right\}, \qquad \rho_c := \frac{1}{1+2\pi}, \qquad k_{11}(\rho) := \sqrt{\frac{\pi^2}{(1-\rho)^2} + 132}. \tag{5}$$

The middle-ratio target is $P < W$ on $\mathcal{R}_{\mathrm{c}}$.

For $0 < z < K$, direct differentiation gives

$$-A'(z) = \frac{1}{\pi}\begin{cases} \arccos\dfrac{z}{K} - \arccos\dfrac{z}{\rho K}, & 0 < z < \rho K, \\ \arccos\dfrac{z}{K}, & \rho K \le z < K. \end{cases} \tag{6}$$

In particular, A is strictly decreasing from T to 0. Let

$$R : [0, T] \longrightarrow [0, K]$$

be its decreasing inverse, and set

$$F(t) := R(t)^2, \qquad g(t) := -F'(t), \qquad \tau := A(\rho K) = K G_1(\rho). \tag{7}$$

The derivative in (6) increases in magnitude on $(0, \rho K)$ and decreases in magnitude on $(\rho K, K)$. The inverse curvature has the related, but slightly stronger, property

$$g \text{ decreases on } (0, \tau), \qquad g \text{ increases on } (\tau, T). \tag{8}$$

Indeed, on the outer branch, if $z = R(t)$ then

$$g(t) = \frac{2\pi z}{\arccos(z/K)};$$

the quotient increases with z, while z decreases with t. On the inner branch, put

$$\delta(z) := \arccos\frac{z}{K} - \arccos\frac{z}{\rho K}.$$

Differentiation with respect to the intermediate ratio gives

$$\frac{\delta(z)}{z} = \int_\rho^1 \frac{ds}{s\sqrt{s^2K^2 - z^2}}.$$

This integral increases with z. Thus $z/\delta(z)$ decreases with z, and $g(t) = 2\pi z/\delta(z)$ increases as t increases and z decreases. The endpoint and interface data are

$$F(\tau) = \rho^2 K^2, \qquad g(\tau) = \frac{2\pi\rho K}{\arccos\rho}, \qquad g(T) = \frac{2\pi\rho K}{1-\rho}. \tag{9}$$

Here and below $g(T)$ is the one-sided limit. Near $t = 0$, one has $g(t) = O(t^{-1/3})$, so all improper Stieltjes boundary terms used below vanish.

2 Exact layer cake and angular rounding

Let

$$x_n := n - \frac{1}{4}, \qquad M(r) := \#\left\{\ell \in \mathbb{N}_0 : \ell + \frac{1}{2} < r\right\} = \max\left\{0, \left\lceil r - \frac{1}{2}\right\rceil\right\}. \tag{10}$$

Lemma 1 (Exact layer cake). *For every $0 < \rho < 1$ and $K > 0$,*

$$P(\rho, K) = \sum_{\substack{n \ge 1 \\ x_n < T}} M(R(x_n))^2, \qquad W(\rho, K) = \int_0^T F(t)\, dt. \tag{11}$$

Both identities retain the strict convention at every radial and angular wall.

Proof. For a fixed half-integer ν,

$$\left[A(\nu)+\frac{1}{4}\right]_{<}=\#\{n\geq 1: x_n<A(\nu)\}.$$

Since A and R are decreasing, $x_n<A(\nu)$ is equivalent to $\nu<R(x_n)$. Interchanging the two finite sums and using

$$\sum_{\ell=0}^{m-1}(2\ell+1)=m^2$$

proves the first identity. At $R(x_n)=m+1/2$, the strict angular count is m, exactly as specified by M.

For the second identity, the area formula for a decreasing inverse gives

$$\int_0^T R(t)^2\,dt=\int_0^K 2zA(z)\,dz.$$

The last integral is (4). $\square$

For every active n, abbreviate

$$r_n:=R(x_n),\qquad m_n:=M(r_n),\qquad \alpha_n:=r_n-m_n+\frac{1}{2}. \tag{12}$$

The strict convention implies

$$m_n-\frac{1}{2}<r_n\leq m_n+\frac{1}{2},\qquad 0<\alpha_n\leq 1. \tag{13}$$

This includes $m_n=0$. Define the angular rounding defect

$$\mathcal{E}_{\text{ang}}(\rho,K):=\sum_{x_n<T}(m_n^2-r_n^2). \tag{14}$$

The defect of a single radial layer is exactly

$$m_n^2-r_n^2=(1-2\alpha_n)r_n+\left(\alpha_n-\frac{1}{2}\right)^2. \tag{15}$$

Unlike the one-sided estimate $m_n^2-r_n^2<r_n+1/4$, formula (15) retains the cancellation when $\alpha_n>1/2$.

Combining (11) and (14) gives the first exact two-term decomposition:

$$W-P=\mathcal{D}_{\text{rad}}-\mathcal{E}_{\text{ang}},\qquad \mathcal{D}_{\text{rad}}:=\int_0^T F(t)\,dt-\sum_{x_n<T}F(x_n). \tag{16}$$

3 The radial deficit with retained positive remainders

Define the shifted counting function and its two primitives by

$$C(t):=\#\{n\geq 1: x_n<t\},\quad w(t):=t-C(t),\quad \mathfrak{W}(t):=\int_0^t w(s)\,ds,\quad \Psi(t):=\mathfrak{W}(t)-\frac{t}{4}. \tag{17}$$

On $0 \le t \le 3/4$,

$$\Psi(t) = \frac{1}{2}\left(t - \frac{1}{4}\right)^2 - \frac{1}{32}. \tag{18}$$

For $t = x_n + u$, $0 \le u \le 1$,

$$\Psi(t) = \frac{1}{2}\left(u - \frac{1}{2}\right)^2 - \frac{1}{32}. \tag{19}$$

Consequently

$$\mathfrak{W}(t) \ge 0, \qquad -\frac{1}{32} \le \Psi(t) \le \frac{3}{32} \quad (t \ge 0). \tag{20}$$

Proposition 2 (Exact retained-remainder identity)**.** *For every $0 < \rho < 1$ and $K > 0$, put*

$$\mathcal{R}_{\text{dec}} := -\int_{(0,\tau)} \mathfrak{W}(t)\, dg(t), \tag{21}$$

$$\mathcal{R}_{\text{osc}} := \left(\Psi(T) + \frac{1}{32}\right) g(T) + \int_{(\tau,T)} \left(\frac{3}{32} - \Psi(t)\right) dg(t). \tag{22}$$

Then both remainders are nonnegative, and

$$\begin{aligned} \mathcal{D}_{\text{rad}} = {} & \frac{F(\tau)}{4} - \frac{g(T)}{8} + g(\tau)\left(\frac{\tau}{4} + \frac{3}{32}\right) \\ & + \mathcal{R}_{\text{dec}} + \mathcal{R}_{\text{osc}}. \end{aligned} \tag{23}$$

Therefore

$$W - P = \mathcal{B}(\rho, K) + \mathcal{R}_{\text{dec}} + \mathcal{R}_{\text{osc}} - \mathcal{E}_{\text{ang}}, \tag{24}$$

where

$$\mathcal{B}(\rho, K) := \frac{\rho^2 K^2}{4} - \frac{\pi \rho K}{4(1-\rho)} + \frac{2\pi\rho K}{\arccos \rho}\left(\frac{K G_1(\rho)}{4} + \frac{3}{32}\right). \tag{25}$$

Proof. Distributionally, $dw = dt - dC$. Stieltjes integration by parts, using $F(T) = 0$ and $w(0) = 0$, gives

$$\mathcal{D}_{\text{rad}} = \int_0^T w(t) g(t)\, dt. \tag{26}$$

The improper boundary term at 0 vanishes because $\mathfrak{W}(t) = t^2/2$ there and $g(t) = O(t^{-1/3})$.

On $(0, \tau)$, integrate with $d\mathfrak{W} = w\, dt$:

$$\int_0^\tau wg = \mathfrak{W}(\tau) g(\tau) - \int_{(0,\tau)} \mathfrak{W}\, dg. \tag{27}$$

On (τ, T), use $w = 1/4 + \Psi'$ and $\int_\tau^T g = F(\tau)$:

$$\int_\tau^T wg = \frac{F(\tau)}{4} + \Psi(T) g(T) - \Psi(\tau) g(\tau) - \int_{(\tau,T)} \Psi\, dg. \tag{28}$$

Since $\mathfrak{W}(\tau) = \tau/4 + \Psi(\tau)$, adding (27) and (28) cancels the uncontrolled interface value $\Psi(\tau)$.

Now add and subtract the extremal sawtooth values in (20). The exact identity

$$\begin{aligned} & \Psi(T) g(T) - \int_{(\tau,T)} \Psi\, dg \\ & \quad = -\frac{g(T)}{8} + \frac{3g(\tau)}{32} + \left(\Psi(T) + \frac{1}{32}\right) g(T) + \int_{(\tau,T)} \left(\frac{3}{32} - \Psi\right) dg \end{aligned}$$

gives (23). By (8), $-dg$ is a positive measure on $(0, \tau)$ and dg is a positive measure on (τ, T). Equations (20), (21), and (22) therefore prove nonnegativity of both remainders. Finally substitute (9) and $\tau = KG_1(\rho)$, and then use (16). □

4 A smooth lower bound for the decreasing-branch remainder

The term $\mathcal{R}_{\mathrm{dec}}$ is quantitatively important on the middle-ratio region; discarding it loses too much. It has the following elementary smooth minorant.

Lemma 3 (A global primitive minorant)**.** *For $t \geq 0$, define*

$$L(t) := \frac{t^2}{2(1+2t)}, \qquad L'(t) = \frac{t(1+t)}{(1+2t)^2}. \tag{29}$$

Then

$$0 \leq L(t) \leq \mathfrak{W}(t) \qquad (t \geq 0). \tag{30}$$

Proof. For $0 \leq t \leq 3/4$, one has $\mathfrak{W}(t) = t^2/2 \geq L(t)$. For $t \geq 3/4$, (20) gives

$$\mathfrak{W}(t) \geq \frac{t}{4} - \frac{1}{32}.$$

Moreover,

$$\frac{t}{4} - \frac{1}{32} - \frac{t^2}{2(1+2t)} = \frac{6t-1}{32(1+2t)} \geq 0.$$

This proves (30). □

Define

$$\mathcal{R}_*(\rho, K) := -\int_{(0,\tau)} L(t)\, dg(t). \tag{31}$$

Because $-dg$ is positive on the decreasing branch,

$$0 \leq \mathcal{R}_* \leq \mathcal{R}_{\mathrm{dec}}. \tag{32}$$

Integration by parts is legitimate at 0 because $L(t)g(t) = O(t^{5/3})$. Hence

$$\mathcal{R}_* = -L(\tau)g(\tau) + \int_0^\tau L'(t)g(t)\, dt. \tag{33}$$

On the outer branch $t = A(z)$, one has $g(t)\, dt = -2z\, dz$. Therefore

$$\mathcal{R}_* = -\frac{\tau^2}{2(1+2\tau)}g(\tau) + \int_{\rho K}^{K} 2z\, \frac{A(z)(1+A(z))}{(1+2A(z))^2}\, dz. \tag{34}$$

This expression is smooth at every point with $0 < \rho < 1$ and $K > 0$ and has no floor or wall discontinuity.

Corollary 4 (A concrete sufficient inequality)**.** *For every $0 < \rho < 1$ and $K > 0$, define*

$$\begin{aligned} \mathcal{H}(\rho, K) := {} & \frac{\rho^2 K^2}{4} - \frac{\pi \rho K}{4(1-\rho)} \\ & + \frac{2\pi\rho K}{\arccos \rho}\left(\frac{\tau}{4(1+2\tau)} + \frac{3}{32}\right) \\ & + \int_{\rho K}^{K} 2z\, \frac{A(z)(1+A(z))}{(1+2A(z))^2}\, dz, \qquad \tau = KG_1(\rho). \end{aligned} \tag{35}$$

If
$$\mathcal{E}_{\mathrm{ang}}(\rho, K) < \mathcal{H}(\rho, K), \tag{36}$$
then $P(\rho, K) < W(\rho, K)$.

Proof. By (25), (34), and
$$\frac{\tau}{4} - \frac{\tau^2}{2(1+2\tau)} = \frac{\tau}{4(1+2\tau)},$$
one has $\mathcal{H} = \mathcal{B} + \mathcal{R}_*$. Now use (24), (32), and $\mathcal{R}_{\mathrm{osc}} \geq 0$. □

5 The reduced angular inequality

The preceding calculation reduces the global structural argument to the following statement on the upward-closed middle-ratio region:
$$\boxed{\sum_{\substack{n\geq 1\\ n-1/4<T}} \big(M(R(n-1/4))^2 - R(n-1/4)^2\big) < \mathcal{H}(\rho, K)} \quad ((\rho, K) \in \mathcal{R}_{\mathrm{c}}). \tag{37}$$

This is the angular inequality passed to the companion block estimate. Everything to its right is the smooth action expression (35), and everything to its left is an explicitly centered angular rounding error. In particular, writing $r_n = m_n - 1/2 + \alpha_n$ gives its exact summands as (15); replacing them by $r_n + 1/4$ destroys the cancellation and is too coarse for the middle-ratio range.

The remaining discontinuities are transparent. For a fixed n, the summand is smooth until r_n crosses a half-integer. At $r_n = m + 1/2$, strict counting assigns $M(r_n) = m$; immediately on the side $r_n > m + 1/2$, the summand jumps upward by $2m + 1$. These are exactly the original proxy walls, now separated from the radial Stieltjes deficit.

There is a natural second summation problem behind (37). Put
$$e(r) := M(r)^2 - r^2, \qquad q(r) := -A'(r), \qquad h(t) := e(R(t)). \tag{38}$$
On every complete angular cell ($m \geq 1$),
$$\int_{m-1/2}^{m+1/2} (m^2 - r^2)\, dr = -\frac{1}{12}. \tag{39}$$
Thus the continuous comparison
$$\int_0^T h(t)\, dt = \int_0^K e(r)q(r)\, dr$$
has a favorable cell-average drift. Passing from this integral to the shifted samples $h(n - 1/4)$ produces the genuine double-sawtooth correlation. Its atoms contain the proxy-wall jumps and cannot simply be discarded by sign. In particular, if an angular wall coincides with a radial sample, Stieltjes integration by parts must retain the common-jump term dictated by the strict convention. The companion angular-block lemma pairs those jump terms with the positive quantity already retained in $\mathcal{H}$ by using the single-peaked shape of q from (6).

Remark 5 (Status). All identities and inequalities through Corollary 4 are analytic and require no finite computation. Equation (37) is the remaining angular inequality in this structural module; the companion angular-block and scalar-positivity modules prove it throughout $\mathcal{R}_{\mathrm{c}}$. Thus no upper cutoff in K is used or needed in the structural implication.

Global Angular Blocks for the Middle-Ratio Shell Proxy

Global analytic angular module

Abstract

The global structural reduction for the spherical-shell proxy leaves an angular rounding defect $\mathcal{E}_{\text{ang}}$ and a smooth positive payment $\mathcal{H}$. This note controls the defect for every $0 < \rho < 1$ and $K > 0$ with at least one active radial layer. First, an exact Stieltjes formula is given with the strict common-jump convention made explicit. The negative mean of every complete angular cell is combined with the single-peaked shape of $q = -A'$ to obtain continuous block estimates. A paired radial block argument then controls the full shifted sample sum, including all common jumps, by

$$\mathcal{E}_{\text{ang}} < \frac{(1-\rho^2)K^2}{6} - \frac{N}{2},$$

where N is the number of active shifted radial layers. On the upward-closed middle-ratio region $\rho_c \le \rho \le 39/50,\ K \ge k_{11}(\rho)$, one has $N \ge 1$. The global middle-ratio theorem is therefore reduced to the smooth scalar inequality (28), which is established in the companion scalar-positivity module.

1 Definitions and the target payment

For $a > 0$ and $z \ge 0$, define

$$G_a(z) = \begin{cases} \dfrac{1}{\pi}\left(\sqrt{a^2 - z^2} - z \arccos \dfrac{z}{a}\right), & 0 \le z < a, \\ 0, & z \ge a. \end{cases} \tag{1}$$

Fix $0 < \rho < 1$ and $K > 0$, and put

$$A(z) := G_K(z) - G_{\rho K}(z), \qquad T := A(0) = \frac{(1-\rho)K}{\pi}. \tag{2}$$

The function A decreases strictly from T to 0. Denote its decreasing inverse by $R : [0, T] \to [0, K]$, and set

$$F(t) := R(t)^2, \qquad q(z) := -A'(z), \qquad \tau := A(\rho K) = K G_1(\rho). \tag{3}$$

Direct differentiation gives

$$q(z) = \frac{1}{\pi}\begin{cases} \arccos \dfrac{z}{K} - \arccos \dfrac{z}{\rho K}, & 0 < z < \rho K, \\ \arccos \dfrac{z}{K}, & \rho K \le z < K. \end{cases} \tag{4}$$

Consequently

$$0 \le q(z) \le \frac{1}{2}, \qquad q \text{ increases on } (0, \rho K), \qquad q \text{ decreases on } (\rho K, K). \tag{5}$$

Let

$$x_n := n - \frac{1}{4}, \qquad M(r) := \max\left\{0, \left\lceil r - \frac{1}{2} \right\rceil\right\}, \qquad N := \#\{n \geq 1 : x_n < T\}. \tag{6}$$

For $1 \leq n \leq N$, write

$$r_n := R(x_n), \qquad m_n := M(r_n), \tag{7}$$

and define the exact angular rounding defect

$$\mathcal{E}_{\mathrm{ang}}(\rho, K) := \sum_{n=1}^{N} (m_n^2 - r_n^2). \tag{8}$$

The strict convention is built into M: at $r = m + 1/2$ one has $M(r) = m$, not $m + 1$.

The smooth payment obtained from the retained radial Stieltjes remainder is

$$\begin{aligned} \mathcal{H}(\rho, K) := {} & \frac{\rho^2 K^2}{4} - \frac{\pi \rho K}{4(1-\rho)} \\ & + \frac{2\pi\rho K}{\arccos \rho}\left(\frac{\tau}{4(1+2\tau)} + \frac{3}{32}\right) \\ & + \int_{\rho K}^{K} 2z \, \frac{A(z)(1+A(z))}{(1+2A(z))^2} \, dz. \end{aligned} \tag{9}$$

The preceding structural reduction proves

$$\mathcal{E}_{\mathrm{ang}} < \mathcal{H} \quad \Longrightarrow \quad P(\rho, K) < W(\rho, K). \tag{10}$$

This note derives a smooth upper bound for the left side.

2 Complete angular cells and the single peak

Define the angular sawtooth error

$$e(r) := M(r)^2 - r^2. \tag{11}$$

For $m \geq 1$, the strict m-cell is

$$I_m = (m - 1/2, m + 1/2], \qquad e(r) = m^2 - r^2 \quad (r \in I_m).$$

Endpoint values do not affect the following Lebesgue integrals. The positive and negative half-cell masses are

$$P_m := \int_{m-1/2}^{m} (m^2 - r^2)\, dr = \frac{m}{4} - \frac{1}{24}, \tag{12}$$

$$Q_m := -\int_{m}^{m+1/2} (m^2 - r^2)\, dr = \frac{m}{4} + \frac{1}{24}. \tag{13}$$

In particular,

$$\int_{m-1/2}^{m+1/2} e(r)\, dr = P_m - Q_m = -\frac{1}{12}. \tag{14}$$

Lemma 1 (Weighted complete-cell bounds)**.** *Let* $m \geq 1$ *and suppose the complete cell* $[m - 1/2, m + 1/2]$ *lies in* $[0, K]$.

1. If $m+1/2 \le \rho K$, then

$$\int_{m-1/2}^{m+1/2} e(r)q(r)\,dr \le -\frac{q(m)}{12} \le 0. \tag{15}$$

2. If $m-1/2 \ge \rho K$, then

$$\int_{m-1/2}^{m+1/2} e(r)q(r)\,dr \le P_m q(m-1/2) - Q_m q(m+1/2). \tag{16}$$

3. On the unique cell, if any, whose interior contains ρK,

$$\int_{m-1/2}^{m+1/2} e(r)q(r)\,dr \le P_m q(\rho K). \tag{17}$$

The initial partial cell $0 \le r \le 1/2$ contributes nonpositively.

Proof. On the left half of a cell $e \ge 0$, and on the right half $e \le 0$. If q is increasing, then $q(r) \le q(m)$ on the positive half and $q(r) \ge q(m)$ on the negative half. Equations (12)–(14) give (15).

If q is decreasing, then $q(r) \le q(m-1/2)$ on the positive half and $q(r) \ge q(m+1/2)$ on the negative half. This gives (16). On the peak cell, discard the negative half and use $q \le q(\rho K)$ on the positive half. Finally, $M(r) = 0$ and $e(r) = -r^2$ for $0 \le r \le 1/2$. □

Thus the continuous angular defect has a favorable analytic block structure:

$$\int_0^T e(R(t))\,dt = \int_0^K e(r)q(r)\,dr. \tag{18}$$

All complete cells below the peak are nonpositive, and the decreasing side is controlled by the endpoint differences in (16). The difficulty is passing from the integral in (18) to the shifted samples in (8); the next subsection records that passage with all common jumps included.

3 Exact common-jump convention

Let

$$h(t) := e(R(t)), \qquad C(t) := \#\{n : 0 < x_n \le t,\ x_n < T\}, \qquad p(t) := C(t) - t. \tag{19}$$

Choose the right-continuous versions of C, p, and h. Right continuity of h is exactly the strict angular convention: when $R(t) = m + 1/2$, the post-jump value uses $M = m$. The atomic measure dC therefore samples the correct value even when a radial point x_n is also an angular wall.

Lemma 2 (Euler identity with common jumps)**.** *With the preceding representatives,*

$$\mathcal{E}_{\mathrm{ang}} = \int_0^T h(t)\,dt - \int_{(0,T]} p(t^-)\,dh(t). \tag{20}$$

At a common radial–angular wall the second integral uses the left value $p(t^-) = -3/4$; no extra or omitted wall term is permitted.

Proof. The measure dC has a unit atom at every active x_n, so

$$\mathcal{E}_{\text{ang}} = \int_{(0,T)} h(t)\, dC(t).$$

Since $dC = dt + dp$,

$$\mathcal{E}_{\text{ang}} = \int_0^T h(t)\, dt + \int_{(0,T]} h(t)\, dp(t).$$

For right-continuous functions of bounded variation, the product formula in the form

$$\int_{(0,T]} h(t)\, dp(t) = [hp]_0^T - \int_{(0,T]} p(t^-)\, dh(t)$$

already contains the product of simultaneous jumps. Here $p(0) = 0$ and $h(T) = e(0) = 0$, so the boundary term vanishes. This proves (20). If $t = x_n$, then $C(t^-) = n - 1$ and hence $p(t^-) = n - 1 - (n - 1/4) = -3/4$, including a common angular wall. □

Equation (20) explains why the negative means in (14) cannot be used without a jump audit. The second term contains both the continuous rise inside each angular cell and the downward atom at its strict lower wall. A direct sign deletion would be invalid. The paired estimate below controls the integral and all its atoms simultaneously.

4 A paired radial-block estimate

The cap $q \le 1/2$ in (5) forces a useful separation: one unit of radial action spans at least two units of angular radius. The shift $x_n = n - 1/4$ leaves a left block of length 3/4, enough to pay the worst angular rounding in that layer.

Lemma 3 (One-layer paired block)**.** *For every active n,*

$$r_n \le \frac{4}{3}\int_{n-1}^{x_n} R(t)\, dt - \frac{3}{4}, \tag{21}$$

and

$$m_n^2 - r_n^2 < r_n + \frac{1}{4}. \tag{22}$$

Both statements remain valid, with the displayed strictness, at every radial–angular common wall.

Proof. Fix $t \in [n-1, x_n]$ and put $z = R(t) \ge r_n$. Since $A(r_n) = x_n$, (5) gives

$$x_n - t = A(r_n) - A(z) = \int_{r_n}^{z} q(s)\, ds \le \frac{z - r_n}{2}.$$

Hence

$$R(t) \ge r_n + 2(x_n - t).$$

Integration over the interval of length 3/4 yields

$$\int_{n-1}^{x_n} R(t)\, dt \ge \frac{3}{4} r_n + \frac{9}{16},$$

which is (21).

By strict counting, $m_n < r_n + 1/2$ even when r_n is a half-integer: at $r_n = m + 1/2$ the value is $m_n = m$. Squaring gives

$$m_n^2 < (r_n + 1/2)^2 = r_n^2 + r_n + 1/4,$$

and proves (22). □

Proposition 4 (Global angular block bound)**.** *For every $0 < \rho < 1$ and every $K > 0$ with $N \geq 1$,*

$$\boxed{\mathcal{E}_{\text{ang}}(\rho, K) < \frac{(1-\rho^2)K^2}{6} - \frac{N}{2}.} \tag{23}$$

In particular,

$$\mathcal{E}_{\text{ang}}(\rho, K) < \frac{(1-\rho^2)K^2}{6} - \frac{1}{2}. \tag{24}$$

Proof. The left blocks $[n-1, x_n]$, $1 \leq n \leq N$, are disjoint subsets of $[0, T]$. Summing (21) and then (22) gives

$$\begin{aligned}
\mathcal{E}_{\text{ang}} &< \sum_{n=1}^{N} \left(r_n + \frac{1}{4} \right) \\
&\leq \frac{4}{3} \sum_{n=1}^{N} \int_{n-1}^{x_n} R(t)\, dt - \frac{N}{2} \\
&\leq \frac{4}{3} \int_0^T R(t)\, dt - \frac{N}{2}.
\end{aligned}$$

The area identity for decreasing inverses gives

$$\int_0^T R(t)\, dt = \int_0^K A(z)\, dz.$$

Scaling $z = as$ in (1) and using

$$\int_0^1 \sqrt{1-s^2}\, ds = \frac{\pi}{4}, \qquad \int_0^1 s \arccos s\, ds = \frac{\pi}{8},$$

shows that

$$\int_0^a G_a(z)\, dz = \frac{a^2}{8}.$$

Consequently

$$\int_0^T R(t)\, dt = \frac{(1-\rho^2)K^2}{8},$$

which proves (23). Equation (24) follows from $N \geq 1$. $\square$

5 The upward-closed middle-ratio scalar inequality

Set

$$\rho_c := \frac{1}{1+2\pi}, \qquad k_{11}(\rho) := \sqrt{\frac{\pi^2}{(1-\rho)^2} + 132}, \tag{25}$$

and consider the upward-closed middle-ratio region

$$\mathcal{R}_c := \left\{ (\rho, K) : \rho_c \leq \rho \leq \frac{39}{50}, \quad K \geq k_{11}(\rho) \right\}. \tag{26}$$

On this set,

$$T \geq \frac{(1-\rho)k_{11}(\rho)}{\pi} = \sqrt{1 + \frac{132(1-\rho)^2}{\pi^2}} > 1, \tag{27}$$

so $N \geq 1$. Proposition 4 and (10) reduce the global middle-ratio theorem to the single inequality

$$\boxed{\mathcal{S}(\rho, K) := \mathcal{H}(\rho, K) - \frac{(1-\rho^2)K^2}{6} + \frac{1}{2} > 0} \qquad ((\rho, K) \in \mathcal{R}_{\mathrm{c}}). \tag{28}$$

Everything in $\mathcal{S}$ is smooth at every finite point of $\mathcal{R}_{\mathrm{c}}$; the floor functions and all radial–angular common jumps have disappeared. Substituting (9) gives the explicit form

$$\begin{aligned}\mathcal{S}(\rho, K) = {} & \frac{(5\rho^2 - 2)K^2}{12} - \frac{\pi \rho K}{4(1-\rho)} + \frac{1}{2} \\ & + \frac{2\pi\rho K}{\arccos \rho}\left(\frac{KG_1(\rho)}{4(1 + 2KG_1(\rho))} + \frac{3}{32}\right) \\ & + \int_{\rho K}^{K} 2z \, \frac{A(z)(1 + A(z))}{(1 + 2A(z))^2} \, dz.\end{aligned} \tag{29}$$

The elementary identity

$$\frac{u(1+u)}{(1+2u)^2} = \frac{1}{4} - \frac{1}{4(1+2u)^2} \tag{30}$$

turns the same scalar gap into the more useful loss form

$$\begin{aligned}\mathcal{S}(\rho, K) = {} & \frac{(1+2\rho^2)K^2}{12} - \frac{1}{2}\int_{\rho K}^{K} \frac{z}{(1+2A(z))^2} \, dz \\ & - \frac{\pi\rho K}{4(1-\rho)} + \frac{1}{2} + \frac{2\pi\rho K}{\arccos\rho}\left(\frac{KG_1(\rho)}{4(1+2KG_1(\rho))} + \frac{3}{32}\right).\end{aligned} \tag{31}$$

Thus the only non-endpoint quantity still requiring control is the positive boundary-layer loss integral in (31).

Remark 5 (Status and next step). The complete-cell bounds, the common-jump formula, and the global estimate (23) are proved analytically. The first remaining step within this module is the smooth scalar inequality (28). The companion scalar-positivity module proves it on all of $\mathcal{R}_{\mathrm{c}}$ by a uniform upper bound for the boundary-layer loss in (31) and strict positivity on the base curve. No upper cutoff in K occurs in the angular estimate or in the verification that $N \geq 1$.

Purely Analytic Positivity of the Upward-Closed Scalar Gap

Global middle-ratio analytic module

Abstract

The angular-block reduction for the middle-ratio spherical-shell proxy leaves one smooth scalar inequality. This note proves that inequality on the upward-closed region $\rho_c \le \rho \le 39/50$, $K \ge k_{11}(\rho)$ without interval arithmetic or a finite numerical certificate. The loss integral is first differentiated with respect to the spectral cutoff. A turning-point lower bound and an exact beta integral show that the scalar gap is strictly increasing in that cutoff for every $K > 12$. At the base curve, a rational secant majorant gives a uniform analytic estimate for the loss. The remaining one-variable inequality is then reduced to positivity of a degree-nine polynomial; all of its Bernstein coefficients on one rational interval are displayed and are positive. Thus there is no upper-frequency cutoff in the scalar-positivity theorem.

1 The scalar gap

For $0 \le s \le 1$, put

$$G_1(s) := \frac{1}{\pi}\left(\sqrt{1-s^2} - s\arccos s\right). \tag{1}$$

Fix

$$\rho_c := \frac{1}{1+2\pi}, \qquad k_{11}(\rho) := \sqrt{\frac{\pi^2}{(1-\rho)^2} + 132}, \tag{2}$$

and consider

$$\mathcal{R}_{\mathrm{c}} := \left\{(\rho, K) : \rho_c \le \rho \le \frac{39}{50}, \quad K \ge k_{11}(\rho)\right\}. \tag{3}$$

The scalar gap left by the angular-block estimate is

$$\begin{aligned}\mathcal{S}(\rho, K) := \frac{(1+2\rho^2)K^2}{12} - \mathcal{J}(\rho, K) - \frac{\pi\rho K}{4(1-\rho)} + \frac{1}{2}\\ + \frac{2\pi\rho K}{\arccos\rho}\left(\frac{KG_1(\rho)}{4(1+2KG_1(\rho))} + \frac{3}{32}\right),\end{aligned} \tag{4}$$

where

$$\mathcal{J}(\rho, K) := \frac{K^2}{2}\int_\rho^1 \frac{s}{(1+2KG_1(s))^2}\,ds. \tag{5}$$

Theorem 1 (Upward-closed scalar positivity)**.** *For every $\rho_c \le \rho \le 39/50$ and every $K \ge k_{11}(\rho)$,*

$$\boxed{\mathcal{S}(\rho, K) > 0.} \tag{6}$$

We use only the elementary rational estimates

$$\frac{157}{50} < \pi < \frac{22}{7}, \qquad \frac{140}{99} < \sqrt{2}, \qquad \frac{5}{3} < \sqrt{3}. \tag{7}$$

The two square-root estimates follow immediately by squaring; the displayed bounds for π are the classical Archimedean bounds (with the lower one weakened to 157/50).

2 Strict monotonicity in the spectral cutoff

The identity

$$G_1(s) = \frac{1}{\pi}\int_s^1 \arccos u \, du \tag{8}$$

and $1 - \cos v \le v^2/2$ imply

$$G_1(s) \ge \frac{2\sqrt{2}}{3\pi}(1-s)^{3/2}. \tag{9}$$

Indeed, $\arccos u \ge \sqrt{2(1-u)}$, and integration gives (9).

Lemma 2 (Derivative of the boundary-layer loss)**.** *For every fixed* $0 < \rho < 1$ *and every* $K > 0$,

$$\partial_K \mathcal{J}(\rho, K) = K \int_\rho^1 \frac{s}{(1 + 2KG_1(s))^3}\, ds. \tag{10}$$

Moreover, if $K > 12$*, then*

$$\int_\rho^1 \frac{s}{(1+2KG_1(s))^3}\, ds < \frac{88}{567}. \tag{11}$$

Proof. Differentiating (5) under the integral sign gives

$$\begin{aligned}
\partial_K \mathcal{J} &= K \int_\rho^1 \frac{s}{(1+2KG_1(s))^2}\, ds - 2K^2 \int_\rho^1 \frac{sG_1(s)}{(1+2KG_1(s))^3}\, ds \\
&= K \int_\rho^1 \frac{s}{(1+2KG_1(s))^3}\, ds,
\end{aligned}$$

which proves (10).

Set $a = 4\sqrt{2}/(3\pi)$. By (9), the change of variable $t = 1 - s$, and $s \le 1$,

$$\begin{aligned}
\int_\rho^1 \frac{s\, ds}{(1+2KG_1(s))^3} &\le \int_0^\infty \frac{dt}{(1 + aKt^{3/2})^3} \\
&= \frac{8\pi}{27\sqrt{3}}(aK)^{-2/3}.
\end{aligned} \tag{12}$$

The last equality is the beta integral

$$\frac{2}{3}(aK)^{-2/3} B\left(\frac{2}{3}, \frac{7}{3}\right) = \frac{8\pi}{27\sqrt{3}}(aK)^{-2/3}.$$

At $K = 12$, the cube of the last member of (12) is

$$\frac{\pi^5}{59049\sqrt{3}}.$$

Using (7),

$$\frac{\pi^5}{59049\sqrt{3}} < \frac{5153632}{1654060905} < \left(\frac{88}{567}\right)^3,$$

where the second strict inequality is the exact identity

$$\left(\frac{88}{567}\right)^3 - \frac{5153632}{1654060905} = \frac{27812576}{44659644435} > 0.$$

The right side of (12) decreases with K, proving (11). $\square$

Proposition 3 (Monotonicity of the scalar gap)**.** *For every $\rho_c \le \rho \le 39/50$ and every $K \ge k_{11}(\rho)$,*

$$\partial_K \mathcal{S}(\rho, K) > 0. \tag{13}$$

Consequently,

$$\mathcal{S}(\rho, K) \ge \mathcal{S}\big(\rho, k_{11}(\rho)\big). \tag{14}$$

Proof. Write

$$\theta := \arccos\rho, \qquad y := KG_1(\rho).$$

The derivative of the final line of (4) is

$$\frac{2\pi\rho}{\theta}\left(\frac{y(1+y)}{2(1+2y)^2} + \frac{3}{32}\right) \ge \frac{3\rho}{8}, \tag{15}$$

because $\theta \le \pi/2$. Also

$$-\frac{\pi\rho}{4(1-\rho)} > -\frac{11\rho}{14(1-\rho)}.$$

Every $K \ge k_{11}(\rho)$ in the stated ratio interval is greater than 12; this also follows from the stronger estimate established in Lemma 4 below. Lemma 2 therefore gives

$$\begin{aligned}
\partial_K \mathcal{S} &> K\left(\frac{1+2\rho^2}{6} - \frac{88}{567}\right) + \frac{3\rho}{8} - \frac{11\rho}{14(1-\rho)} \\
&= K\left(\frac{13}{1134} + \frac{\rho^2}{3}\right) + \frac{3\rho}{8} - \frac{11\rho}{14(1-\rho)} \\
&> \frac{26}{189} + 4\rho^2 + \frac{3\rho}{8} - \frac{11\rho}{14(1-\rho)}.
\end{aligned} \tag{16}$$

It remains to check the sign of the last expression. Multiplication by $1-\rho$ produces

$$p(\rho) := (1-\rho)\left(\frac{26}{189} + 4\rho^2 + \frac{3\rho}{8}\right) - \frac{11\rho}{14}. \tag{17}$$

The bound $\pi < 22/7$ gives $\rho_c > 7/51$. On the larger rational interval $[7/51, 39/50]$, set

$$u := \frac{\rho - 7/51}{39/50 - 7/51} \in [0,1].$$

In the degree-three Bernstein basis $B_{j,3}(u) = \binom{3}{j}u^j(1-u)^{3-j}$, exact expansion gives

$$p(\rho) = \sum_{j=0}^{3} b_j B_{j,3}(u), \tag{18}$$

with

j	b_j
0	$\frac{335005}{2785671}$
1	$\frac{32947387}{196635600}$
2	$\frac{281783183}{578340000}$
3	$\frac{231517}{13500000}$

(19)

All four coefficients are positive. Since the Bernstein basis functions are nonnegative and sum to one, $p(\rho) > 0$. Equation (16) proves (13), and (14) follows by integration in K. □

3 An exact rational upper bound for the loss

Lemma 4 (Uniform loss bound)**.** *For every $\rho_c \leq \rho \leq 39/50$ and every $K \geq k_{11}(\rho)$,*

$$K > \frac{241}{20} \tag{20}$$

and

$$\int_\rho^1 \frac{x}{(1+2KG_1(x))^2}\,dx < \frac{17}{100}. \tag{21}$$

Proof. Since $\rho \geq \rho_c$,

$$\frac{\pi}{1-\rho} \geq \frac{\pi}{1-\rho_c} = \pi + \frac{1}{2} > \frac{91}{25}.$$

Consequently

$$K^2 \geq k_{11}(\rho)^2 > 132 + \left(\frac{91}{25}\right)^2 = \left(\frac{241}{20}\right)^2 + \frac{471}{10000},$$

which proves (20). The rational bounds in (7) now give

$$\frac{4\sqrt{2}K}{3\pi} > \frac{4(140/99)(241/20)}{3(22/7)} = \frac{23618}{3267} = \frac{36}{5} + \frac{478}{16335}. \tag{22}$$

Put $y = 1 - x$. Equations (9) and (22) yield

$$\begin{aligned}\int_\rho^1 \frac{x\,dx}{(1+2KG_1(x))^2} &< \int_0^{1-\rho} \frac{1-y}{(1+(36/5)y^{3/2})^2}\,dy \\ &\leq \int_0^1 \frac{1-y}{(1+(36/5)y^{3/2})^2}\,dy.\end{aligned} \tag{23}$$

We bound the last integral by exact rational arithmetic. Set

$$c := \frac{36}{5}, \qquad \phi(z) := \frac{1}{(1+z)^2},$$

and use the rational grid

$$(t_i)_{i=0}^{10} = \left(0, \frac{1}{4}, \frac{5}{16}, \frac{3}{8}, \frac{7}{16}, \frac{1}{2}, \frac{9}{16}, \frac{5}{8}, \frac{3}{4}, \frac{7}{8}, 1\right). \tag{24}$$

For $z_i = ct_i^3$, let

$$\beta_i := \frac{\phi(z_{i+1}) - \phi(z_i)}{z_{i+1} - z_i}, \qquad \alpha_i := \phi(z_i) - \beta_i z_i.$$

Because $\phi''(z) = 6(1+z)^{-4} > 0$, its secant satisfies

$$\phi(z) \le \alpha_i + \beta_i z \qquad (z_i \le z \le z_{i+1}).$$

After $y = t^2$, the integral in (23) is

$$\int_0^1 2t(1-t^2)\phi(ct^3)\,dt.$$

Thus it is at most

$$Q := \sum_{i=0}^{9} \big(F_i(t_{i+1}) - F_i(t_i)\big), \tag{25}$$

where the exact polynomial antiderivative is

$$F_i(t) := \alpha_i \left(t^2 - \frac{t^4}{2}\right) + 2c\beta_i \left(\frac{t^5}{5} - \frac{t^7}{7}\right). \tag{26}$$

Every number in (25) is rational. Substitution of the grid (24) gives

$$Q = \frac{13057672380614461540641104348359387792539833731785583268887}{77064462602711372906638041909709692886169218555152043745280}, \tag{27}$$

and direct subtraction gives

$$\frac{17}{100} - Q = \frac{21643130923235926743681388145629999054466710986445549053}{385322313013556864533190209548548464430846092757602187264 00} > 0. \tag{28}$$

Equations (23)–(28) prove (21). $\square$

4 The base curve

Fix ρ and now set

$$t := \sqrt{1-\rho}, \qquad k := k_{11}(\rho) = \sqrt{132 + \frac{\pi^2}{t^4}}, \qquad \rho = 1 - t^2. \tag{29}$$

The middle-ratio range lies strictly inside the rational interval

$$\frac{7}{15} < t < \frac{14}{15}. \tag{30}$$

Indeed,

$$t^2 \ge \frac{11}{50} > \left(\frac{7}{15}\right)^2,$$

while $\rho \ge \rho_c > 7/51$ gives

$$t^2 = 1 - \rho < \frac{44}{51} < \left(\frac{14}{15}\right)^2.$$

Lemma 5 (Analytic endpoint payment)**.** *For every $\rho_c \le \rho \le 39/50$, let t and k be as in (29), and let*

$$C(\rho) := \frac{100\rho^2 - 1}{600} \tag{31}$$

and

$$E(t) := \frac{11}{14t^2} - \frac{14}{15(1+2t)} - \frac{3}{8t}. \tag{32}$$

Then $C(\rho) > 0$, $E(t) > 0$, and

$$\mathcal{S}(\rho, k) > C(\rho)k^2 - \rho E(t)k + \frac{1}{2}. \tag{33}$$

Proof. First, $\rho \ge \rho_c > 7/51 > 1/10$, so $C(\rho) > 0$. Lemma 4 and

$$\frac{1+2\rho^2}{12} - \frac{17}{200} = \frac{100\rho^2 - 1}{600}$$

show that the first two terms in (4) are strictly greater than $C(\rho)k^2$.

Set $\theta = \arccos\rho$ and $Y = kG_1(\rho)$. For $0 \le t \le 1$,

$$\theta = \arccos(1 - t^2) = 2\arcsin\frac{t}{\sqrt{2}} \le \frac{\pi t}{2}. \tag{34}$$

To see the last inequality, use convexity of arcsin on $[0, 1/\sqrt{2}]$: its graph lies below the chord joining $(0, 0)$ to $(1/\sqrt{2}, \pi/4)$. Furthermore, (9) and $k > \pi/t^2$ give

$$Y > \frac{2\sqrt{2}}{3}t > \frac{14}{15}t. \tag{35}$$

Since $Y \mapsto Y/(1 + 2Y)$ is increasing, (34)–(35) imply

$$\begin{aligned} \frac{2\pi\rho k}{\theta}\left(\frac{Y}{4(1+2Y)} + \frac{3}{32}\right) \\ > \rho k\left(\frac{14}{15(1+2t)} + \frac{3}{8t}\right). \end{aligned} \tag{36}$$

Also

$$-\frac{\pi\rho k}{4t^2} > -\frac{11\rho k}{14t^2}.$$

Combining these inequalities with (4) proves (33).

Finally,

$$840t^2(1+2t)E(t) = 660 + 1005t - 1414t^2. \tag{37}$$

The quadratic on the right is concave, so its minimum on $[7/15, 14/15]$ occurs at an endpoint. Its endpoint values are

$$\frac{184739}{225}, \qquad \frac{82406}{225},$$

respectively. Both are positive, proving $E(t) > 0$. □

Proposition 6 (Rational positivity on the base curve)**.** *For every $\rho_c \le \rho \le 39/50$,*

$$\mathcal{S}\big(\rho, k_{11}(\rho)\big) > 0. \tag{38}$$

Proof. The elementary inequality

$$\sqrt{x^2+132} < x + \frac{66}{x} \qquad (x>0)$$

and (7) give

$$k^2 > 132 + \frac{(157/50)^2}{t^4}, \tag{39}$$

$$k < \frac{22}{7t^2} + \frac{3300t^2}{157}. \tag{40}$$

Since $C(\rho) > 0$ and $\rho E(t) > 0$, Lemma 5 yields

$$\mathcal{S}(\rho, k) > R(t), \tag{41}$$

where

$$\begin{aligned} R(t) := C(1-t^2)&\left(132 + \frac{(157/50)^2}{t^4}\right) \\ &- (1-t^2)E(t)\left(\frac{22}{7t^2} + \frac{3300t^2}{157}\right) + \frac{1}{2}. \end{aligned} \tag{42}$$

It remains only exact polynomial algebra. Put

$$H(t) := t^4(1+2t)R(t).$$

Using (37), its factorized form is

$$\begin{aligned} H(t) = C(1-t^2)(1+2t)&\left(132t^4 + \left(\frac{157}{50}\right)^2\right) \\ &- \frac{(1-t^2)(660+1005t-1414t^2)}{840}\left(\frac{22}{7} + \frac{3300}{157}t^4\right) \\ &+ \frac{1}{2}t^4(1+2t). \end{aligned} \tag{43}$$

For an independently checkable expansion,

$$\begin{aligned} H(t) = &-\frac{20642567}{24500000} - \frac{6205067}{12250000}t + \frac{547983}{122500}t^2 - \frac{1033727}{367500}t^3 \\ &+ \frac{34911551}{16485000}t^4 + \frac{187093801}{8242500}t^5 + \frac{8679}{1099}t^6 \\ &- \frac{138149}{2198}t^7 - \frac{2101}{157}t^8 + 44t^9. \end{aligned} \tag{44}$$

Set

$$u := \frac{t-7/15}{7/15} \in [0,1].$$

In the degree-nine Bernstein basis,

$$H(t) = \sum_{j=0}^{9} d_j B_{j,9}(u), \qquad B_{j,9}(u) = \binom{9}{j} u^j (1-u)^{9-j}, \tag{45}$$

where exact conversion of (44) gives

$$
\begin{array}{c|c}
j & d_j \\
\hline
0 & \frac{65408801807437}{9463832437500000} \\
1 & \frac{299549381760793}{1135659892500000} \\
2 & \frac{1590469980572077 3}{28391497312500000} \\
3 & \frac{276560525659002 1}{3154610812500000} \\
4 & \frac{33203943051258761}{28391497312500000} \\
5 & \frac{38636221171331747}{28391497312500000} \\
6 & \frac{2538003206646073}{1892766487500000} \\
7 & \frac{28758289199580211}{28391497312500000} \\
8 & \frac{12920786461478803}{28391497312500000} \\
9 & \frac{3428732855971679}{9463832437500000}
\end{array}
\tag{46}
$$

Every d_j is positive. The Bernstein basis functions are nonnegative and sum to one; hence $H(t) > 0$ throughout $[7/15, 14/15]$. Since $t^4(1+2t) > 0$, this proves $R(t) > 0$. Equation (41) proves (38). $\square$

Proof of Theorem 1. Proposition 3 reduces every point of the upward-closed middle-ratio region to the base curve $K = k_{11}(\rho)$. Proposition 6 proves strict positivity on that curve. Therefore $\mathcal{S}(\rho, K) > 0$ throughout $\mathcal{R}_c$, with no upper restriction on K. $\square$

Remark 7 (Nature of the certificate). The rational secants and Bernstein expansions above are finite exact algebra, not interval arithmetic: convexity proves the functional majorant on each secant interval, and positivity follows from explicitly displayed rational coefficients. No floating-point evaluation, sampled minimum, or numerical root isolation is used as a premise.

Purely Analytic Closure of the Final Spherical-Shell Residual

Abstract

This note assembles the all-frequency retained-remainder estimate after the exact owner subtraction. It proves that the final spherical-shell residual is empty without invoking an interval certificate, an executable truth table, an aggregate tail argument, or a frequency split. The structural, angular-block, and scalar-positivity modules cover one closed superset of the entire residual.

1 Notation and analytic inputs

Let

$$A_{\rho,1} := \{x \in \mathbb{R}^3 : \rho < |x| < 1\}, \qquad W(\rho, K) := \frac{2(1-\rho^3)K^3}{9\pi},$$

and let $N_D(A_{\rho,1}, K^2)$ count Dirichlet eigenvalues strictly below K^2. Put

$$\rho_c := \frac{1}{1+2\pi}, \qquad k_{11}(\rho) := \sqrt{\frac{\pi^2}{(1-\rho)^2} + 132}.$$

The preceding analytic owner lemmas and the exact set subtraction give

$$\mathcal{D}_{20} = \left\{(\rho, K) : \rho_c \le \rho < \frac{39}{50}, \quad k_{11}(\rho) < K < U(\rho)\right\}. \tag{1}$$

On this interval $U(\rho) = K_0(\rho)$, although the formula for the upper wall will not be needed below.

We use the shell action

$$G_a(z) := \frac{1}{\pi}\left(\sqrt{a^2 - z^2} - z \arccos\frac{z}{a}\right) \quad (0 \le z < a), \qquad G_a(z) := 0 \quad (z \ge a),$$

$$A_{\rho,K}(z) := G_K(z) - G_{\rho K}(z),$$

and the strict lattice proxy

$$P(\rho, K) := \sum_{\substack{\nu=\ell+1/2<K \\ \ell\in\mathbb{N}_0}} 2\nu \, \#\left\{n \in \mathbb{N} : n < A_{\rho,K}(\nu) + \frac{1}{4}\right\}. \tag{2}$$

The analytic Bessel-phase reduction proved earlier in the main argument is

$$N_D(A_{\rho,1}, K^2) \le P(\rho, K). \tag{3}$$

Put

$$M_c(\rho, K) := \frac{(1-\rho^2)K^2}{6} - \frac{1}{2}.$$

Parts VII–IX of the detailed analytic supplement establish, respectively, the following three inputs. The retained-remainder module gives the structural implication

$$\mathcal{E}_{\mathrm{ang}}(\rho, K) < \mathcal{H}(\rho, K) \quad \Longrightarrow \quad P(\rho, K) < W(\rho, K). \tag{4}$$

The paired angular-block module gives, on $\rho_c \le \rho \le 39/50$ and $K \ge k_{11}(\rho)$,

$$\mathcal{E}_{\mathrm{ang}}(\rho, K) < M_c(\rho, K). \tag{5}$$

The scalar-positivity module gives on the same unbounded domain

$$\mathcal{H}(\rho, K) - M_c(\rho, K) > 0. \tag{6}$$

The all-frequency scope in (6) is part of that module: its scalar gap has positive K-derivative whenever $K > 12$, one has $k_{11}(\rho) > 241/20 > 12$, and the gap is strictly positive on the base curve $K = k_{11}(\rho)$. Thus the proof integrates upward from the base curve to every finite $K \ge k_{11}(\rho)$; it uses no upper-frequency bound.

2 All-frequency retained-remainder closure

Theorem 1 (All-frequency middle-ratio theorem)**.** *For*

$$\rho_c \le \rho \le \frac{39}{50}, \qquad K \ge k_{11}(\rho),$$

one has

$$N_D(A_{\rho,1}, K^2) < W(\rho, K).$$

Proof. Equations (5) and (6) give the strict chain

$$\mathcal{E}_{\mathrm{ang}}(\rho, K) < M_c(\rho, K) < \mathcal{H}(\rho, K).$$

The structural implication (4) therefore gives $P(\rho, K) < W(\rho, K)$, and the phase reduction (3) completes the proof. Strictness includes both closed ratio faces, the base face $K = k_{11}(\rho)$, and every common radial–angular proxy wall. ☐

3 Direct ownership of the exact residual

Proposition 2 (Analytic closure of $\mathcal{D}_{20}$)**.** *Define the closed all-frequency theorem domain*

$$\mathcal{R}_{\mathrm{rr}} := \left\{ (\rho, K) : \rho_c \le \rho \le \frac{39}{50}, \quad K \ge k_{11}(\rho) \right\}.$$

Then $\mathcal{D}_{20} \subset \mathcal{R}_{\mathrm{rr}}$, the strict Pólya inequality holds throughout $\mathcal{D}_{20}$, and the successor residual is empty.

Proof. If $(\rho, K) \in \mathcal{D}_{20}$, then (1) gives

$$\rho_c \le \rho < \frac{39}{50}, \qquad k_{11}(\rho) < K < U(\rho).$$

After weakening the strict upper ratio and lower frequency inequalities, these conditions place (ρ, K) in $\mathcal{R}_{\mathrm{rr}}$. Therefore Theorem 1 proves every point of $\mathcal{D}_{20}$, and

$$\mathcal{D}_{20} \setminus \mathcal{R}_{\mathrm{rr}} = \varnothing.$$

The already owned faces $\rho = 39/50$, $K = k_{11}(\rho)$, and $K = U(\rho)$ remain outside $\mathcal{D}_{20}$, even though the stronger theorem also covers the first two. The face $\rho = \rho_c$ remains in $\mathcal{D}_{20}$ whenever both strict frequency inequalities hold. No artificial frequency face is present. ☐

Logical dependency. After Parts I–VI identify $\mathcal{D}_{20}$, the closure uses only the structural, angular-block, and scalar-positivity modules in Parts VII–IX. It does not use an aggregate-tail module, a frequency split, the former 10,580-box compact certificate, the 1,286-box aggregate certificate, or the 63-state executable subtraction table.